

# Universe24: an integer cellular automaton read through clicks, the formulas it shares with nature and the ones it adds

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## Abstract

Universe24 is the clicks of one line of integer arithmetic: a cubic lattice, two integer levels and a remainder per part of a field, stepped by one recurrence from a Node's present, past and six neighbours, exactly reversible in the integers, covariant under the cube's 48. The lattice is read only through declared counting regions; a click, the only measurement, is the meeting of the forward field with the transition read backward. From it follow, in lattice form, the Klein-Gordon equation with Schrödinger's slow limit, Planck's relation, and, under one assumption fixed by two measurements, the exponential weak-field metric,  $\gamma = \beta = \mathbf{1}$ , and, with no number put in, frame dragging's form  $\mathbf{g}_{0j} = -\mathbf{4V}_j$  for a plane configuration of one rotation sense, the coefficient  $\mathbf{4}$  the clock's and the ruler's exponents, antimatter anti-dragged and positronium undragged its predictions. The two slits' row computed from the band before the run, **284.7** quanta in the counting window, is read by the run's clicks at **285**. Bell's and Greenberger-Horne-Zeilinger correlations reach Tsirelson's  $\mathbf{2\sqrt{2}}$  under the declared counting rule. The lattice's own terms are predictions where unread and bounds where read: a black-hole shadow radius  $\mathbf{2em}$  against the Event Horizon Telescope; light's gravitational source at  $\mathbf{E/c^2}$ ; light undragged by a rotating mass. Its misses are named, frame dragging read on bodies and a decelerating cosmology among them. Universe24 is one world of a method that builds others; its claim is that it helps compute things in nature, not that nature is such a lattice.

**Keywords:** integer cellular automaton, reversible lattice dynamics, quantum measurement, Bell inequalities, weak-field gravity on a lattice

# 1 Introduction

## 1.1 The question and the claim

This paper is about one method of computing physics. A cubic lattice of Nodes is stepped by one local rule in whole numbers, Rule3, which is exactly reversible. The lattice is read only through clicks, the reports of declared counting regions, which the model calls NodeDetectors. The title names the parts. The integer cellular automaton is the lattice stepped by Rule3. Universe24 is its clicks, written by the NodeDetectors from the fields of its declared universe. The formulas it shares with nature are the known formulas reached from the one line in lattice form: the Klein-Gordon equation with Schrödinger's slow limit, Planck's relation in the band's form (the adiabatic invariant [1]) and, under one named assumption, the exponential weak-field metric of gravity (Section 5.1). The formulas it adds are the lattice's own terms, which depart from the continuum or have no counterpart there (Section 5.2). The paper has one claim: that Universe24 helps compute things in nature. Its content is what the model computes of nature, set against measurement, each quantity with its status and its number, the hits and the misses alike. The method is how it computes, and not the claim. The paper does not claim that nature is such a lattice.

The criterion is the same throughout: a derived, click-fenced formula that disagrees with a measurement beyond its stated order refutes the model at that formula. Section 8 lists the refutations found. Universe24 is one world built by the paper's method: two declared universes, the two slits' (the one run, read by its NodeDetectors) and Bell's (no run), and the forms of Sections 4 and 5, derived from the law's lines and computed from no run. Other worlds closer to nature can be built by the same method. Building one needs the calculations named in Section 8.3, which this work does not do.

The paper starts from one statement and never contradicts it. Universe24 is the meeting of the past with the future; a click is that meeting, the only measurement there is in the model, and in it only clicks are seen (assumption, the paper's first statement; clicks; the key of Section 1.3). A Node holds a field's present and its past, two integer levels. The rule makes the field's future from these two levels and from the six neighbouring Nodes. Run forward from the source Node of its initial data, the split events are the future, every path at once. Read backward from a click that happened, by the same rule, they are the past, and through two slits both of them pass through both gaps. A click is where the two meet at a declared NodeDetector. Which meeting happens is the only thing the model does not determine, the only non-local element of the description (Section 2.5). In the model, what looks like a wave is the future. What looks like a particle is the present, the click. The track is the past.

In the model, whoever sees is itself a cluster of bound states on the lattice. A Node reads its six Ports and nothing else, and beyond the cluster's tail nothing reaches it except what a NodeDetector reports, one quantum credited at a meeting. What stands on the lattice between clicks is not a fact of Universe24 but its possibilities; a fact is a realised event, a quantum credited at a click. So the paper has three words and keeps them apart. The lattice is the lattice stepped by Rule3; it computes. The NodeDetector is the gate between the lattice and Universe24: it reads the lattice and

writes the clicks that happened. Universe24 is those clicks, the world of the model. A lattice reading, a level, a support or a count at a Node, is a diagnostic of the computation and is labelled so. Rule3 is a bijection of the levels (Theorem 1), so no level is an irreversible record and nothing in the lattice is an event; a measurement is an event, one whole quantum credited at a declared region and never two, which only the NodeDetector's draw makes, the one map of the model that is not a bijection, so a measurement of the model is a NodeDetector's click and nothing else, and what an instrument of nature reports is set beside it as such a click by declaration (assumption, the paper's first statement; theorem for the bijection; clicks). Only a click pattern is compared with a measured pattern of nature (assumption, the paper's first statement; clicks). A deterministic, reversible integer lattice read only through clicks gives the statistics of one quantum computed here, one quantum per click, the two slits' pattern and Bell's values, from a wave on the lattice and a declared counting rule, with Born's rule put in and with the Hilbert space beyond one quantum, spin and the many-body atom not reached (derived for the wave and the interference, S.1, S.23; exclusivity derived under the counting rule's one draw, the NodeDetector's declaration, S.46; an assumption, fitted, for the counting rule, S.10, S.38; an assumption for its draw and for the partner's conditional draw of Section 7; the misses by name, Section 8; clicks). The backward reading fixes no number that the forward counting rule does not: the joint energy is computed from the forward configurations alone (Section 2.5).

## 1.2 What the paper adds, and its precursors

The paper adds three things, each with its status in its sentence and in the claims list of the supplementary material. The first is the method: one line of integer arithmetic stepped reversibly on a lattice and read only through clicks, with measurement as the meeting, at a NodeDetector, of the future with the past, the NodeDetector being the only place of the random draw. The second is the forms it reaches. These are the continuum's known formulas in lattice form from that one line, gravity's vector sector among them under the one assumption, each with its inputs and its status (the list of Section 5.1), and the lattice's own terms, which depart from the continuum or have no counterpart there, a prediction where no measurement has read them and bounded where one has (the list of Section 5.2). The third is the method of claim: every statement carries its status and its fence, every input and every miss is named, and the derivations stand in Online Resource 1 and the law's document in the deposit (Statements and Declarations). Among the forms, one result is this paper's own, and read on bodies it is a finding against nature. From the one assumption applied to a moving source, with no number put in, the coefficient 4 being the sum of the clock's and the ruler's exponents, follows frame dragging's form  $g_{0j} = -4U v_{s,j} = -4V_j$  ( $U$  the potential, the clock's share,  $\mathbf{v}_s$  the source's velocity and  $\mathbf{V} = U\mathbf{v}_s$  the odd lines' level) for a plane configuration of one rotation sense; the opposite sense is dragged the opposite way and a one-part real line not at all, so a body is dragged by its plane configurations' signed share of its inertia, about one half for nature's matter, against Gravity Probe B and the Lense-Thirring satellite tests, with  $\alpha_1 = 8f - 8$  for a body of share  $f$ ,  $-4$  at one half, against its bound, where the write without the odd Link part gives  $\alpha_1 = -8$  and no drag (derived under the one assumption and the band's

own Lorentz form at long wavelength and small gap, S.21, to first order in  $\mathbf{V}$ , slow matter; the body's share  $Z/A$  under the proton hypothesis of Section 4.1; clicks at nature's gap; Section 8.1; S.56 (e) to (g)).

The precursors are these, one line each. The physical world has been read as a discrete computation [2] and as reversible cellular automata [3]; the present step is a second-order one in integers with one Euclidean division, its precursor the bit-reversible integer leapfrog [4], here with the remainder kept. Elze's integer-valued cellular automata have Schrödinger's equation as their continuum limit [5]; here the step is one Euclidean division whose remainder stays at the Node, nonlinear in the integers, with two levels and a remainder and no stored complex amplitude, and with the click layer and its one draw. Bialynicki-Birula gave unitary cellular automata for the Weyl, Dirac and Maxwell equations on a lattice [6].

The detailed paragraphs of this subsection stand in Online Resource 1, Section B.0.

Two answers stay. Feynman's bound on a local classical computer [7] holds, with the excess placed in the declared counting rule of the click and not in the lattice (Section 7). A lattice's Lorentz violation at order one unless its terms are tuned [8][9] is answered in part: the model has no loops, an absence listed among the gaps, and the band's factors close with the gap, at a pair the declared integer bound does not reach (Section 8.1).

### 1.3 How to read the paper

Every claim of the paper carries its status in the parenthesis that closes its sentence: the status word first, then its fence, then its pointers. The claims list at the front of the supplementary material (Online Resource 1) holds every claim with its status, its fence, what it rests on and its derivation. The status is one of eight words, all of which the paper and its supplement use:

- *theorem*, proved for every lattice under the conditions its statement names.
- *derived*, obtained from theorems by argument.
- *computed*, a number obtained from the rule's lines by derivation or by the one run of Rule3 the paper reports (Section 6).
- *assumption*, a line held by decision and named where it is used.
- *hypothesis*, a line outside the model, under its own name.
- *experiment*, a number the model does not fix, with the experiment named that would fix it, the paper performing none.
- *fitted*, an input chosen with a known result in view.
- *inspiration*, a direction named in the conjectures of Online Resource 1, Section B.12 (Section 9), with no result claimed.

Beside them, the sentences and the claims list use thirteen more words. Calibration is a number taken from nature. Declaration is a universe's own number or a NodeDetector's own rule, held by decision as an assumption is. Definition marks a definition. Untested means that no measurement has read the quantity. No line means that the law has no line that gives the quantity: nothing is derived and nothing contradicted, and the row names what a line would have to give. Bounded means that a measurement bounds the form and confirms nothing. Open means that the paper decides nothing about the item and claims nothing of it. Named means that the assumption or the input

is stated in the sentence that uses it. Conditional on a line means that the result holds under that line and is not claimed without it. A postulate is one of the stated postulates of the list in Online Resource 1, C.1, an assumption. A prediction is a lattice term no measurement has read. By construction and no test marks an identity of the definitions that no measurement can refute. A comparison is a measurement set beside the model's number, a supporting finding and not a proof. Every formula carries its fence. *Lattice* marks a reading of the lattice's own numbers that no NodeDetector reports, a diagnostic. *Clicks* marks a formula for what a NodeDetector reports, the only kind that can be set beside nature.

A bare “computed” is a number from Rule3's lines, the algebra evaluated, by hand or numerically, with no click drawn, as each sentence says. “Computed, a run” names a fact of the one run of Rule3's integer line the paper reports, the two slits (Section 6). The algebraic steps of every derivation are in the supplementary material (Online Resource 1), Derivations S.1 to S.56, cited as (S.*n*). The derivations and the passages the paper's and the supplement's statuses cite beyond them stand in the same supplement, Section C, cited as (C.*n*). Every declaration and derivation the paper uses stands in this article and Online Resource 1; the law's document in the deposit (Statements and Declarations) is described and not relied on. Every sentence that says what a thing is, light, mass, an atom, gravity, is a sentence about the model and its declared universes, never about nature. The theorems are theorems about the rule, and every comparison with nature is a supporting finding and not a proof. The five coined words of the paper, Node (with its Links and Ports), NodeDetector, Rule3, click and pace, are defined at their first use. A large language model was used as a tool throughout the work, as the Statements and Declarations state.

The detailed paragraphs of Sections 1.2, 2.5.1, 4.1, 4.2, 4.3, 4.4, 5.2, 6, 7, 8 and 9 stand in Online Resource 1, Section B, whole; the result sentences and their marks stay here.

## 2 The model and the rule

The physical location is a *Node*, and its complete local information is its NodeState. For every field and every part, the NodeState holds two integer levels, the present and the past, and one remainder. A part is one level pair of a field: a potential field's time and axis parts, a bound state's own levels, a pair field's two lines. The potential fields' levels are the potential level a Node holds for the other fields to read. Nothing else is kept at a Node, and every count, current, charge or stress is a reading computed from these numbers when it is needed. Nodes are joined by *Links* through six directional *Ports* per Node, and the cubic array of Nodes is the lattice. A *field* is one kind of physical value with its rational pair and its shape, and its *configuration* is its levels over the lattice. A *pace* is the step factor of a Node (its clock) or of a Link (its ruler). A *NodeDetector* is the model's counter, a connected region of Nodes declared in the universe's declaration, and its *click* is one quantum credited to it, the only measurement.

## 2.1 The lattice and its group of 48

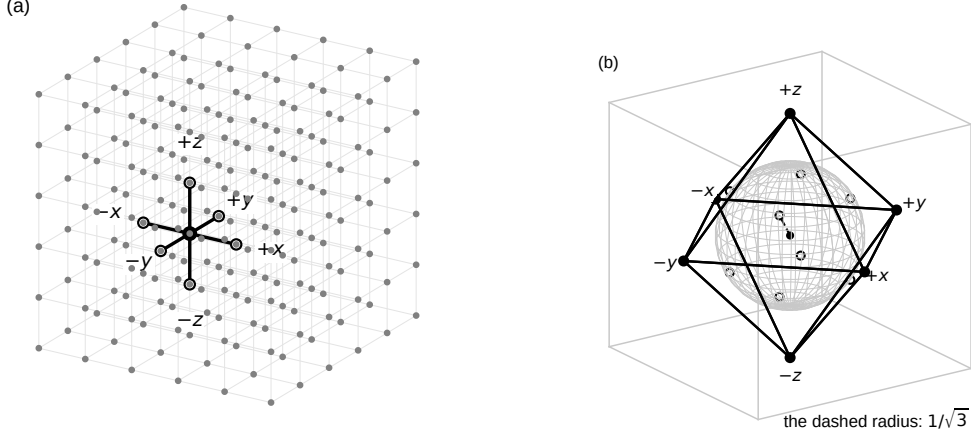
The lattice is a box of  $N_x \times N_y \times N_z$  Nodes, each joined to its six neighbours by Links through its Ports  $\pm x, \pm y, \pm z$  (Fig. 1(a)). An axis is declared periodic, open or folded. Beyond an open face a field's rest is read, 0 for a field without one; a folded axis is one layer, whose two Ports return the Node itself. An inner face is a plane of Nodes declared closed except for declared gaps, a wall with slits. A face may be declared receding (an assumption): the box grows by a layer of zeros whenever a nonzero level reaches the Node next to it, so that the step on such a box is the step on the unbounded lattice, with no cosmological redshift, since nothing stretches.

The symmetry group of the cube has 48 elements: the 24 rotations and, with the reflections, the signed permutations of the three axes. It is  $O_h$ , the group of the octahedron of the Nodes one interval away (Fig. 1(b)), and with the translations it is the space group of the simple cubic lattice. It acts by permuting the Nodes and their Ports and carries every level pair with its Node. Rule3 commutes with it, since its line reads the six Ports symmetrically and names no axis. Under the group the form  $D$  of Eq. (1) is a scalar, even under the time reversal, which exchanges a field's next and before levels, keeps its present, and carries the step to its inverse. The current  $F$  is a vector, odd under the time reversal. The Wronskian  $W$  is a scalar under the cube's 48, odd under the time reversal and under the plane's reflection. The click is a scalar, naming a region and an interval, never a Node or an axis. No boost, no rotation beyond the cube's 24 and no scaling is in the group, so Lorentz's group and the continuum's rotations are no symmetries of the lattice, and what the paper finds of the moving clock it finds as a property of the clicks (theorem for the covariance and the kinds of the readings; lattice; S.4; the moving clock derived, the second order a definition; clicks; Section 5.1, S.21).

## 2.2 Rule3 and the vacuum band

The paper follows from one line of integer arithmetic, Rule3, and the inputs named beside it: one write, three readings, the one assumption of the forms (Section 2.3), the counting rule of the click and the declarations named where they stand. The line holds at every Node, in every interval and for every field:

|                |                                                                                                                                                                                                                                                                                                                                     |     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Rule3:         | $w a_{\text{next}} + r' = \sum_{j \sim i} R_{ij} \text{arr}_j + S a_{\text{now}} - w a_{\text{before}} + r,$ $0 \leq r' < w = 6 \text{den } \Gamma^2,$ $R_{ij} = 2 \text{num } p_a(i, j)^2 = 2 \text{num } p_i^2 q_{ij}^2 / \Gamma^2,$ $S = 12 \text{den } \Gamma^2 - 12 (\text{den} - \text{num}) p_0^2 - \sum_{j \sim i} R_{ij};$ |     |
| the one write: | a potential field's part at a Node gains<br>$\left( \sum_f w_f q_f(\text{Node}) + r \right) \text{div } E, \text{ the remainder } r \text{ kept there;}$                                                                                                                                                                            | (1) |
| the readings:  | $D_i = \text{now}_i^2 - \text{next}_i \text{before}_i,$ $W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}},$ $F_{ij} = \text{num} (\text{now}_i \text{before}_j - \text{before}_i \text{now}_j).$                                                                           |     |



**Fig. 1** (a) The lattice in space, a cubic array of Nodes, here  $6 \times 6 \times 6$ , each joined to its six neighbours by Links through the Ports  $\pm x, \pm y, \pm z$ ; one Node and its six neighbours, the Nodes one interval away from it, are marked. A change passes between neighbouring Nodes at one Link per interval, the causal bound (assumption, the step's reach; lattice; Section 2.2). Light passes at  $1/\sqrt{3}$  of it (derived; lattice; S.1). (b) The Nodes one interval away form the octahedron  $|x| + |y| + |z| \leq 1$ , whose six vertices are the six Ports; its symmetry group is the cube's,  $O_h$ , the 24 rotations and, with the reflections, the 48 signed permutations of the axes. The inscribed sphere, of radius  $1/\sqrt{3}$ , touches the eight faces on the diagonals of the cube, and that radius is light's speed in Link per interval. Both are drawn from the definitions; no run

In the first line  $a_{\text{now}}$ ,  $a_{\text{before}}$  and  $a_{\text{next}}$  are the field's level at the Node now, one interval before and one interval later. The level of the neighbour  $j$  is  $\text{arr}_j$ , the six Ports summed. The remainder kept at the Node before and after the step is  $r$  and  $r'$ . The Node clock is  $\Gamma$ . The field's pair is  $[\text{num}, \text{den}]$ , with  $\cos \omega_0 = \text{num}/\text{den}$  its rotation per interval at wave number zero. The Node's clock pace is  $p_0$ , its own pace  $p_i$ , and  $p_a(i, j)$  is the pace of the Link through the Port, the Node's pace times the Link's factor  $q_{ij}/\Gamma$ . At the vacuum's paces the two Links of an axis share one  $R_a = 2 \text{ num } \Gamma^2$  and  $S = 0$ , and  $\text{div}$  is floor division. One whole number is formed and divided once by the wall  $w$ . The quotient is the next level, the remainder stays at the Node, nothing is rounded away, and the step back is the same division (Theorem 1).

The second line is the only way a potential field's level is ever written (a field that holds a value for others to read). The sourcing fields' quantities  $q_f$  at the Node, each a form  $D_f$ , a Wronskian  $W_f$  or a stress  $T_{aa,f}$ , are read in the Node's own units (Section 2.3; S.34), weighted by  $w_f$ , summed with the part's one remainder and divided by the part's wall  $E$  (a divisor, not the form  $E$  of Section 3.3),  $E_s T$  for a time part and  $E_s W_c$  for an axis part,  $E_s$  the potential's divisor.  $w_f$  carries the write weight  $k_w$  and the proper factors of S.34 (d); the energy line  $E_s T \text{ num} = k_w \Gamma \text{ den}$  is stated in S.26. The third line is what is read from a configuration and never written: its form  $D_i$ , which sources the potential fields; its Wronskian  $W_i$ , its charge (for a plane field  $\text{re}$  and  $\text{im}$  are its two level pairs); and its current  $F_{ij}$  into the Node  $i$  from its neighbour  $j$ . A NodeDetector reads that current through its boundary in quanta of the field's unit  $W_c = 3 \text{ den } T$ ,  $T$  the quantum's norm. One quantum's inflow is  $W_c \sin \omega_0$  in the

current's unit and its energy  $T \sin \omega_0$  in the quantum's, and a massless field's quantum is  $W_c$  itself.

At the vacuum's paces the line reads  $a_{\text{next}} + a_{\text{before}} = \frac{\text{num}}{3 \text{ den}} \sum_a (\text{arr}_{+a} + \text{arr}_{-a})$ . Its rotation at the wave vector  $\mathbf{k}$  is  $\cos \omega = \frac{\text{num}}{3 \text{ den}} \sum_a \cos k_a$ , with  $\cos \omega_0 = \text{num}/\text{den}$  at wave number zero; along one axis it is Meyer's dispersion relation [10] at  $\theta - \rho = \omega_0$  and  $\tan \theta \tan \rho = 2$ , his  $\tan \theta$  the mass, so the inertia  $3 \tan \omega_0$  below is the lattice's and the three axes' equal share from one pair is the line's own (Section 1.2; Online Resource 1, B.0). For the massless pair  $[1, 1]$  along one axis  $\cos \omega = (2 + \cos k)/3$ , so  $\omega = |k|/\sqrt{3}$  as  $k \rightarrow 0$ , and along the cube's diagonal  $\cos \omega = \cos k$  exactly, with no dispersion (derived; lattice; S.1, S.54). Light's speed at long wavelength is  $1/\sqrt{3}$  Link per interval (derived; lattice; S.1). It lies below the causal bound of one Link per interval (assumption, postulate 4 of Online Resource 1, C.1, the step's reach; lattice). Its group speed along an axis falls at shorter wavelengths as  $1 - 0.083 k^2$  (S.1), by  $1/48$  of that on a face diagonal and not at all on the body diagonal (S.24). In nature's names the vacuum's line is the discrete Klein-Gordon equation (a quantum walk's lattice form of it in [11]), exact but for the remainder term, with  $c_K^2 = \text{num}/(3 \text{ den})$  and  $m^2 = 2(1 - \text{num}/\text{den})$ , the rest rotation the mass, and its slow limit at small wave number is Schrödinger's equation with the inertia  $m^* = 3 \tan \omega_0$  under three named conditions (derived; lattice; S.1, S.53). By construction  $m^* c_m^2 = \omega_0$  for every pair, with  $c_m^2 = \omega_0/(3 \tan \omega_0)$  the band's kinetic scale: this is the model's  $E = mc^2$ , read in three click numbers, nature's its small-gap limit, the departure  $\omega_0/\tan \omega_0$ ,  $1 - \omega_0^2/3$  to the order  $\omega_0^2$  (derived for  $m^*$  from the band,  $c_m$  defined, the identity by construction and no test; clicks; S.18, S.21).  $c_K$  is the discrete equation's coefficient at every wave number and  $c_m$  the band's kinetic scale at rest; they differ at a finite gap and agree at nature's (derived; lattice; S.1, S.18, S.21, S.34 (c)). At the vacuum's pace Rule3 is the leapfrog scheme of the Klein-Gordon equation, exact in the variable  $s = 2 \sin(\omega/2)$ . What the law adds is one rational  $[\text{num}, \text{den}]$  that fixes mass and speed together, an equation that is exact, departures that are predictions with bounds (Sections 5.1 and 5.2), the integers with the remainder, and the paces (derived; lattice; S.1).

### 2.3 The paces and the one assumption

A potential field enters a field's step by changing the field's paces at a Node, the coefficients of Rule3 there,  $c_i$  the potential level the field reads at the Node,  $U = c/\Gamma$  the same in the potential's unit, and  $t_a(i, j)$  the Link's stress (the level the Link holds), one number per Link from the parts at its two ends (S.7; the Link's number the mean of its two ends' readings, S.33):

$$\boxed{p_0 = \Gamma (1 - 1/\Gamma)^c, \quad p_i = \frac{p_0(c_i)^2}{\Gamma}, \quad p_a(i, j) = \frac{p_i q_{ij}}{\Gamma}, \quad q_{ij} = \Gamma - t_a(i, j), \quad c = \sum_{\text{reads}} \text{weight} \times \text{the read field's time level.}} \quad (2)$$

A positive weight makes a hollow, an attraction, and a negative weight a hill; every field reads the potential fields its declaration names at the weights it names, a plane

field also the charge potential, and sources what it reads, so every pair of fields that read each other is reciprocal.

Nothing in Rule3 fixes how a level enters the paces. Beside the line, the model holds that a potential level is a count of units and that every act is measured in the Node's own units,  $hN = 1$  in its units form: the one assumption of the forms (assumption, fitted; the composition derived from it; lattice; S.34). From it the clock composes, and a uniform potential level  $U$  changes only the units, time by the lapse  $N = p_0/\Gamma$  and length by  $h = p_0/p_a$  per Link (S.34). One sentence is not in Rule3: a Link is clocked twice by its reading Node's own clock,  $p_i = p_0^2/\Gamma$ , the ruler going with the clock. The lattice says why twice, since a Link has two ends. It does not say why the reading Node's clock stands at both ends rather than the product of the two ends, which gives  $\beta = -1$  at the small gap (the post-Newtonian  $\beta$  as the orbits read it, Section 5.1).

The sentence is one discrete choice. The one assumption,  $hN = 1$ , is fixed by the bending's measurement,  $4U_b$  for "twice" ( $\gamma = 1$ ),  $U_b$  the potential at the impact parameter  $b$ . The perihelion's 43 arcseconds per century [12] choose the Node's own reading of the Link's pace over the two ends' product ( $\beta = -1$  at the small gap, 72 arcseconds) and then stand as the test of the choice:  $\beta = 1$  as the gap closes,  $\beta_K = 1.25$  at the computed pair, where the band's  $k^4$  term adds 1/8 of the advance to first order, 13 percent by the ray integration on the exact band in a coordinate plane (orientation dependent within a percent) and 8 to 10 in the (111) plane (derived under the one assumption; clicks for  $\gamma$ ,  $\beta$  and the redshift's factor, lattice for the metric's form at the computed pair; S.19). It is the first of three choices made with a known result in view (Section 8.2). Every gravitational form of Section 5.1 is conditional on it: the bending's measurement is its calibration and no test of it, the perihelion the test of the end's choice, Shapiro's delay and the redshift's factor  $f(\omega_0) = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$  (S.15) its consequences, read against Cassini and Pound and Rebka, and the shadow's  $2em$  bounded by measurement and not confirmed.

For a moving source the potential's flux writes three odd Link levels  $V_{ij}$ , the lattice's  $g_{0i}$ , read as a phase by plane (charged) configurations and by no real one. The same assumption holds for a source that moves: the field's count flux sources gravity's three odd lines as its count sources the time line, in the count's unit,  $V_{ij} = -V_{ji}$ , and the Port turns an arriving plane configuration of one rotation sense by the phase  $\theta_{ij} = -\theta_{ji} = 4V_{ij}\omega_0/c^2 = 12V_{ij}\omega_0$  of its field, the opposite sense by its negative,  $c^2 = 1/3$  light's own speed squared on the lattice, the three axes' 3 and no number of nature, the 4 the clock's and the ruler's exponents at  $\gamma = 1$  and no number put in, paid inside Rule3's one division, the odd Link part of the metric, gravity's potential field's three odd parts; a one-part configuration, light among them, carries no phase (assumption, the one assumption for a moving source; the coefficient derived under it and the band's own Lorentz form at long wavelength and small gap, S.21; lattice; S.56). Under it the model's static metric for light is  $ds^2 = -e^{-2U}c^2dt^2 + e^{2U}d\mathbf{x}^2$ ,  $c^2 = 1/3$ ,  $t$  in intervals and  $\mathbf{x}$  in Links, to  $O(U/\Gamma)$  and in the ray limit: the exponential metric [13], Einstein's to the first post-Newtonian order for a static source,  $\gamma = \beta = 1$ , with the frame's parameters derived under the same assumption for a moving source (derived under the one assumption; lattice; S.34 (c); Section 5.1). The clock's rate is  $N$ , exactly in the lattice's variable  $2\sin(\omega/2)$  and in  $\omega$  up to the finite-rotation correction of order

$\omega_0^2/12$ , 6.3 percent at the computed pair as the exact  $f(\omega_0) - 1$  (S.51) and  $\omega_e^2/12$ ,  $\omega_e$  the electron's rest rotation at nature's pair,  $= 1.1 \times 10^{-20}$  at the Large High Altitude Air Shower Observatory's (LHAASO) bound of Section 5.1 (derived under the one assumption; lattice; S.34 (c), S.31 for the clock's rate, S.51, S.24).

From the coefficients, for a line whose two opposite Links of each axis share one coefficient, the remainder-free line of S.1, the rotation of a configuration at a Node with clock pace  $p_0$  and Link paces  $p_a$  is (S.1)

$$\cos \omega = 1 - \left(1 - \frac{\text{num}}{\text{den}}\right) \left(\frac{p_0}{\Gamma}\right)^2 - \frac{\text{num}}{3 \text{ den}} \sum_a \left(\frac{p_a}{\Gamma}\right)^2 (1 - \cos k_a). \quad (3)$$

The mass term scales with the clock's square and the band term with the square of the Link's pace: at  $k = 0$  the rest rotation of every massive band obeys  $1 - \cos \omega_b = (p_0/\Gamma)^2(1 - \cos \omega_0)$ , and along a Link the speed of every wave falls with the Link's pace, so light is delayed and bent by twice the clock's share, its speed at a Node of potential  $U = x$  being  $N^2 = e^{-2x}$  times its vacuum speed, an index  $n = e^{2x}$ , to the unit (derived; lattice; S.2). A pace must lie at or below the guard's edge  $P$ ,  $P^2 = 2 \text{ den } \Gamma^2 \text{ div } (\text{den} + |\text{num}|)$ , the Courant, Friedrichs and Lewy condition [14] of the explicit scheme that Rule3 is at the vacuum's pace, a condition on the declared universe (theorem for the edge; lattice; S.3). Light's vacuum pace sits at the edge at the level 0, where S.3 admits the linearly growing checkerboard mode, and below it at the declared universes' level 60 (computed; lattice; S.3). Under the composed paces no Link closes in the rational paces, since the clock is positive at every level, so a deep well freezes a clock and closes nothing (derived for the closing; assumption for the frozen Node's energy; lattice; S.39). A Node whose Link pace rounds to 0, at  $(\Gamma/2) \ln 2\Gamma$  levels, is a frozen Node, read by its neighbours, reading nothing and dark (derived for the threshold under the one assumption; assumption for the dark Node; lattice; S.3, S.39).

## 2.4 The interval and the clock

A reading writes nothing on the lattice, and a static configuration (now = next = before) has no form and no current, so it carries no quanta and no NodeDetector sees it. Every read takes the values at the interval's start, so the interval's dependency radius is one Link (assumption, the law's line, postulate 4 of Online Resource 1, C.1, the step's reach; lattice). Its four acts, in order, are the read of the paces, Rule3 on every configuration, the readings with each NodeDetector's report, and the write of each potential field; after the four acts, at a realised click, the NodeDetector's one write (Section 2.5). The conservations are per field, at static paces and in the rational line, the integer step within the remainders' term, and the lattice conserves no total, since the quantum share (the field's count in a region, Section 3.4) moves with the paces and no current compensates; the click layer has a count register of its own, one integer per configuration that clicks alone move, and neither layer keeps a total energy (theorem for the step's conservations; assumption for the clicks'; lattice, clicks; S.49).

The line and its clock are fixed: every mend of the model is a field, a potential field, a declaration or a read, never a change of Rule3 (an assumption).

## 2.5 The click: the meeting and the NodeDetector

*Assumption 1* (the click is a meeting, the law and not a hypothesis) A field's configuration is stepped on the lattice in both directions by the one line: forward from the source Node of its initial data, the possibilities, the future; backward from the realised click, what happened, the past, Rule3 at  $-1$ . A click is the meeting of two quanta at a declared NodeDetector: the quantum of the incoming configuration run forward, the future, and the transition of the configuration that waits for it, the click's unit, the past read back from it. Its quantum share is their product at the region, at resonance (Section 4.3): the form  $D = \text{now}^2 - \text{next} \times \text{before}$ , the now against the future times the past. Which meeting is realised is the NodeDetector's random draw, the world's one (a bound state's NodeDetector's own seed), and the lattice draws nothing. The one non-local place is the counting rule's one draw through the source Node, the one declared non-local element, the half-up rounding across regions and the backward reading its two faces. The NodeDetector only asks whether the future and the past met at it.

There is one kind of reader, and the declared universe decides what it holds: with no configuration of its own a NodeDetector reads the field that arrives at it, a bolometer's boundary, as the two slits' screen does; given a configuration of its own, with parts, transitions and a lifetime, it stands for an atom or an ion and emits as well. Algebraically, the step  $S_t$  is a bijection of the state space (Theorem 1), its forward orbit the future and its inverse orbit the past. A NodeDetector  $R$  is a declared region whose boundary Ports and coefficients define  $Q_R$ , the inflow over the counting window, and a click is the pair  $(R, t)$ , one whole quantum credited to  $R$  at the interval  $t$ , drawn with the probability  $s_R / \sum_{R'} s_{R'}$ ,  $s_R = Q_R(x_t) / W_{\text{rec}}$ ,  $x_t$  the state at the interval  $t$ ,  $W_{\text{rec}}$  the configuration's energy per quantum (Section 3.4),  $W_c$  for light and  $W_c \sin \Omega$  for a born quantum,  $\Omega = \omega_e - \omega_g$  the beat of the emitting configuration's excited and ground rotations (assumption for the definition, derived for the precisions; clicks; S.46). The decision is the one map in the model that is not a bijection.

The parts of a configuration are read with the NodeDetector's coefficients, each part paired with the same part through the common source Node, then summed and squared; the two slits are the case with nothing to pair, Bell's check the case through the common source Node (Section 7), the joint energy computed from the forward configurations alone by the linearity of the rule (derived; clicks; S.46). A NodeDetector with Nodes alone counts by the half-up rounding of the total over every region reading the configuration, so one region's count moves with the others' inflows by at most one quantum per window, the one non-local element of the draw of Assumption 1; a bound state weighs each transition by its own transition's quantum share, so its marginal is its own whatever the other bound states' (derived under the counting rule's declaration; clicks; Online Resource 1, C.8 (b)).

*Assumption 2* (the NodeDetector's declaration, its decision and its write) A NodeDetector is one connected region declared in the universe's declaration, two Nodes or more and at least half the wavelength of the field it reports across the beam (4 Links at  $\lambda = 8$  Links), one

Node or more where the declared universe gives it a configuration of its own. Each interval it reads the inflow through its Nodes' boundary Ports, Rule3's currents summed in integers into its counting window, declared in the universe's declaration and never read off the wave, which opens at its last write and closes at its draw. Its click is its report: the window, the NodeDetector, the field and the part, the count before and after, its own clock, never a Node, one unit of the field,  $W_c \sin \omega_0$  of the inflow's energy, being one unit of count. The counting rule credits one quantum to one NodeDetector, drawn with the world's one random draw by the energies of the inflows over every declared NodeDetector of the declared universe, one credit per quantum and never two, the draw deterministic per seed.

After a realised click the NodeDetector writes once, at the one Node of its region that the same draw next picks by the inflow counted through each Node: one whole quantum of the arriving field is absorbed there, its count in the register falls by one, the absorbing configuration's part rises by one, laid in its mode with the count's own scaling,  $A^2 = (N + 1) T / (2 \sin \omega_{\text{part}})$  on each of the part's two lines,  $\omega_{\text{part}}$  the laid part's own rotation (the law's write). Where the arriving field's energy at the drawn Node is at most  $W_{\text{rec}}$  its levels and remainder there go to 0, and a denser field is undepleted, its count moving in the register alone and the deficit recorded in the register. Nothing is scaled by a factor and no number enters, and the change spreads by Rule3 as an event (Section 2.5.1). Backward, to the past, a reading and never a write: the counting rule's reading through the source Node, which chooses which tree is the past. A NodeDetector given a configuration of its own, held by its declaration and never bound by a well, emits as well as takes, one whole quantum of light laid at the drawn Node at  $\Omega = \omega_e - \omega_g$ , drawn per window at the rate, the beat's current over  $W_c$  plus the rate's declared constant (the transition beat labels the emitting transition; the emitted excitation's frequency distribution and its frequency-selective absorption are not derived from this write; assumption for the write; lattice; S.1, S.46).

What produces a click is deterministic, the meeting. Which possibility becomes the click no line of the law gives: the NodeDetector draws a place and a combination, never a value and never a path. The draw is a declared linear congruence of the full period [15]. So Born's rule on the lattice [16] is the proportionality of the weights to the energies, a declaration of the NodeDetector and not a finding of the model, each outcome's frequency within  $2^{-b}$ ,  $b$  the declared bound's exponent, of its weight's share, the draw's uniformity over its full period, and not exactness ( $1.1 \times 10^{-19}$  at the declared  $b = 63$ ; S.46). The successive draws' independence is assumed and not proven, and the scatter  $\sqrt{NP(1-P)}$  is conditioned on it. Exclusivity, one click per quantum, follows from the one draw, the NodeDetector's declaration, with the count conserved (derived; clicks; S.46). It gives antibunching [17], one quantum and one click per configuration between far NodeDetectors; under it no signal passes through the amplitudes (Rule3 is local, Theorem 1) nor through a bound state's clicks (C.8 (b)); a Nodes-only region's count is coupled to the others' inflows by the credit's rounding, the declared non-locality of C.1 (3), by at most one quantum per window,  $1/N$  relative (derived under the counting rule's one draw, the NodeDetector's declaration, for exclusivity; derived for antibunching and the one form; assumption for the draw, the write and the counting rule; clicks; lattice for the write; S.46).

The NodeDetector's resolutions are its own declarations and no uncertainty relation of the model's own: in place the region ( $\Delta y k \geq \pi$  at half a wavelength), in frequency the counting window ( $\Delta \omega \geq 2\pi/n$  over  $n$  intervals) and in count the draw

( $\sqrt{NP(1-P)}$  over  $N$  clicks). The dead time is not derivable, a declaration: the quantum shares credit a region any whole number of quanta in one interval, and  $W_c$  over the inflow's rate is the mean spacing of the clicks and not a least one (declaration; derived under it; clicks; S.8, S.46). The lattice holds only the future and has no arrow, Rule3 being a bijection at every interval; the clicks have the arrow, each forward only, the one act outside the reversal: the direction of time belongs to the clicks and not to the lattice (theorem for the step, the arrow the NodeDetector's declaration; lattice; S.46). A quantum changes field or part only at a click's write, and between clicks every count is constant (the count constant between clicks, and the click the one act of the law that moves a count, derived, S.46; the exchange itself the NodeDetector's declaration; that in nature only a click moves a quantum between fields a hypothesis under its own name; clicks). Light is the charge potential's own configuration, sourced by the Wronskian of the fields that read it, stepped by Rule3 at the paces of its reads and written by the one write: a static rotating bound state writes a static level, Coulomb's field, and radiates nothing, and a breathing one writes a varying source that the charge potential carries away, the spreading events of Section 2.5.1, the light (derived; lattice for the Wronskian and the source, clicks for the far NodeDetector's rate; S.40). No configuration reads its own write of the charge potential.

*The steps up to a click, with two NodeDetectors.* At A a window closes: the draw picks a place and a combination, the write moves one count, A's part down by one and one quantum of light born at the beat, its frequency set by the beat, its wave number by light's band as it spreads, its direction none (the transition beat labels the emitting transition; the emitted excitation's frequency distribution and its frequency-selective absorption are not derived from this write; assumption for the write; lattice; S.1, S.46). From A's Node the quantum spreads by Rule3 as events, a fan of possibilities inside the causal cone at one Link per interval, light's front within it at  $1/\sqrt{3}$ , the wave vector the phase step between neighbours; nothing travels as a point. At B the inflow is read at its Ports and the shares accumulate; the configuration run forward from A meets the transition that B's click reads backward, Rule3 at  $-1$ : the meeting. At the window's close B draws whether and which future continues, the NodeDetector and the combination, under Born's weights of the shares; its write's Node is the draw's next pick. B's write moves one count, the light's down by one and B's part up by one, at the drawn Node; the drawn future is now a past. From B's Node the erasure runs over the rest of the fan (Section 2.5.1). Between the two clicks one count has moved twice and one frequency, set at A, has been carried by Rule3 and read at B, locally at every Link; the one thing not local is B's draw, one per configuration over every NodeDetector that reads it (assumption for the draw, the write, the counting rule, the erasure and the empty wave; derived for the rest; lattice, clicks; S.46).

### 2.5.1 How an event spreads

An event spreads by Rule3: the level splits to the six neighbours along the Links, one Link per interval, the causal bound of the line, which reads the six Ports and nothing farther; light is such spreading events at  $1/\sqrt{3}$  Link per interval (S.1), and a click's erasure runs at the full bound inside its cone. The causal front after  $d$  intervals is the octahedron of the Link distance, and light's front at long wavelength is the sphere of

radius  $d/\sqrt{3}$  inscribed in it; the two touch at the faces' centres, on the four diagonals, where the band is  $\cos \omega = \cos k$  exactly, and along the axes the bound runs  $\sqrt{3}$  ahead of the light (assumption, postulate 4, for the bound; derived for the two fronts; lattice; S.1). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.1.

### 2.5.2 The click's algebra beyond the law

A click is the one act of the law that moves a register's count,  $-1$  for the arriving configuration and  $+1$  for the taker's part, the taker the configuration whose transition the click reads or the quantum leaving the lattice at a declared face, every other act leaving every count unchanged (assumption, the NodeDetector's declaration, for the exchange; every other act leaving the booked count unchanged, by the booking (Section 3.4; S.46); the quantum share moving by the currents and the paces' drift (Theorem 4; S.5, S.6); lattice, clicks). The draw chooses whether and which future continues, a place and a combination, never a value, a path or the time (assumption, the counting rule; clicks). The write at the drawn Node changes the share by the click's identity under a write (the list of Section 5.2) (derived; clicks; S.46). It keeps the charge, balances the frequency by the beat (Section 4.3) and conserves no momentum, the quanta laid with a phase and no wave number (the law's write, an assumption, the NodeDetector's declaration; the charge kept by Lemma A, S.55 (a); lattice, clicks; S.46; Section 8.1). A quantum's momentum is a chain: its frequency is set at its birth click by the beat and its wave number by light's band, its direction none at the one-Node laying of the quantum; it is carried by Rule3, conserved in vacuum and turned in wells by the index with the frequency kept, and read at its death click (derived; lattice; S.2, S.7, S.20).

One proposition stands beyond the law's write of Section 2.5, under its own name, chosen with a known result in view, that nature's clicks conserve momentum event by event [18] (hypothesis, fitted; clicks; no number of the paper's claims rests on it; the results attributed to it below (the recoil, Compton's factor 0.752, 0.778, 0.5015) are its own, conditional on it). Under it the write carries the arriving wave vector as a phase twist of the taker's configuration,  $\delta k_a = \text{sgn}(k_a) (|k_a| \text{div } M)$  per Link along each axis for a taker of count  $M$ , the remainder carried, the wave vector read at the region's Ports from the Link current and no number beyond its own reads (hypothesis, fitted, for the twist; derived under it for the read at the Ports; lattice; S.55 (c)). Derived under it in S.55: the charge kept and the integer wave vector conserved across the click,  $r_a + k_a = M \delta k_a + r'_a$  at every click, the register before and after, the twist written on the taker and the remainder carried (derived under the proposition; lattice; S.55). The recoil  $k^2/(2Mm^*)$  in rotation with  $m^* = 3 \tan \omega_0$  vanishes for a large taker (derived under the proposition; lattice; S.55).

Compton's shift in the non-relativistic regime [19] has the factor  $\omega_0/\tan \omega_0$ , 0.752 at [2, 3] from the exact factor, 0.778 to first order in the gap  $1 - \text{num/den}$ , the beat rule's number (S.55 (f)), and 1 at nature's gap, against 0.5015 at [2, 3] under the click's energy  $T \sin \omega$ , the lattice's own term at a finite gap, where Compton's formula is the Link's definition by construction, no test (derived under the proposition, nature's by construction and no test; lattice at the computed pair; S.55 (f)). In a well of index

$n = e^{2U}$  the wave vector passed per click, the twist, is  $n$  times the vacuum's at the same frequency, Minkowski's form, and the recoil in rotation  $n^2$  times it (derived under the proposition; lattice; S.2, S.55). The paragraph is algebra on the law's definitions; S.55 carries the algebra, its checks and the integer threshold of the twist.

### 3 Exact properties

This section states what holds exactly, each item under the conditions its theorem names, each item proved in S.45, Theorem 4 in S.5 and S.6, the Wronskian's conservation in S.9 and the three exact bands in Section 3.6's own line: the reversibility of the step, the kind of function its acts can form on the infinite lattice at fixed coefficients, the conserved form at static paces with its current, the count as a reading of the configuration under the two conditions on the lattice (S.5, S.6), the Wronskian's conservation under the line without its division (S.9) and the three exact bands.

#### 3.1 Reversibility

Let  $\sigma$  be the direction, +1 forward and -1 backward,  $\text{div}$  and  $\text{mod}$  the floor division and its remainder in  $[0, w)$ , and  $\rho$  the remainder carried in. With  $u = \sigma(\sum_j R_{ij} \text{arr}_j + Sx - wy) + \rho$ ,  $z = \sigma(u \text{ div } w)$  and  $\rho' = u \text{ mod } w$ , the forward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{before}}, r)$  gives  $(z, \rho') = (a_{\text{next}}, r')$ , and the backward call with  $(x, y, \rho) = (a_{\text{now}}, a_{\text{next}}, r')$  gives  $(z, \rho') = (a_{\text{before}}, r)$ .

**Theorem 1** (the direction) *The backward call returns  $(a_{\text{before}}, r)$  exactly, integer for integer; the reversibility of every step needs no second function, the odd lines' source recomputed by the backward step as the paces are (S.56 (b)). (theorem for the step and where the writes change no level; the write an assumption; lattice; S.45).*

A NodeDetector's write is outside the reversal, as the initial data are; the whole evolution goes back,  $N$  intervals forward and  $N$  back returning every level of every field exactly, integer for integer, provided every number an act read at an interval's start is one the state after the interval still holds. On the two slits' universe the run forward and back returns every level exactly, integer for integer, over 40 intervals, the counting window opening at 45, so the check contains no click (computed, a run; lattice; S.23). The condition rules out one read: a coefficient read from a Node's own state across a step cannot be recovered at the step back, so a configuration's rotation sense is never read at a Node, and the charge potential is read by the configuration's shape, a plane or a line (Section 3.5).

#### 3.2 What the acts can form

Rule3's line is one act, and declared coefficients make four special cases of it, which the law also calls acts: the read, the line with its six arrivals; the one-Node step, the arrivals declared 0, which rotates a level by Chebyshev's recurrence; the division,  $R = S = 0$ , Euclid's division with the remainder kept; and the load, a declared integer in the numerator, which is how a count enters a level.

**Theorem 2** (what the acts can form, on the infinite lattice, for one field at fixed coefficients)  
*Every fixed finite composition of the line's four acts with fixed coefficients is an affine form of one field's levels plus a term bounded by a constant of the composition, since a floor is its argument less a fraction in  $[0, 1]$ ; so no product of two of its levels is written by the step on the whole lattice. The products across fields enter through the coefficients the read turns from the held levels: the read multiplies. On a bounded domain every integer function is a finite sum of floors of affine forms, and the theorem says nothing there. (theorem; lattice; S.45, S.56 (c)).*

So every product of two levels is a reading and never a write by the step, and the readings of Eq. (1), the stress and the share are the only products the law takes. A line enters the law only if it passes three tests, generic (one primitive with declared integers, no field's name), vector (an act on the whole state, with no square root and no real number) and local (its own configuration and the six neighbours); a line that needs more is a hypothesis under its own name.

### 3.3 The conserved form

The read coefficient  $R_a(i \rightarrow j) = 2 \text{ num } p_i^2 q_{ij}^2 / \Gamma^2$  carries the Node's factor and the Link's; where the paces differ the Node's factor is not symmetric and the Link's is, so the conserved form carries the weight  $1/p_i^2$  on its Node terms and the Link's factor on its Link terms:

$$E = \sum_i \frac{w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{ now}_i \text{ before}_i}{p_i^2} - 2 \text{ num } \sum_i \sum_{j \sim i} \frac{q_{ij}^2}{\Gamma^2} \text{ now}_i \text{ before}_j. \quad (4)$$

**Theorem 3** (the conserved form) *At constant paces for which a positive Node weight  $d_i$  with  $d_i R_a(i \rightarrow j) = d_j R_a(j \rightarrow i)$  on every Link exists,  $E$  with the weights  $d_i$  is exact under the line without the rounding. Under the law's read coefficient, with  $q_{ij} = q_{ji}$  and  $p_i = p_0(c_i)^2 / \Gamma$ , the weight  $d_i = 1/p_i^2$  cancels the Node's factor on both sides of every Link. So the weighted form, with  $\mathbf{M}$  the matrix of Rule3's step,  $\mathbf{a}_{\text{next}} + \mathbf{a}_{\text{before}} = \mathbf{M} \mathbf{a}_{\text{now}}$ ,  $\mathbf{D} = \text{diag}(1/p_i^2)$  and  $\mathbf{D}\mathbf{M}$  symmetric at every interval, is exact for any static potential levels and static Link stresses, and for a plane configuration under the Port's phase, the weighted coupling Hermitian, the same form exactly (S.56 (a)). Its Link term carries the Link's factor, so the current of  $E$  through the Link  $ij$  is  $2 \text{ num } (q_{ij}^2 / \Gamma^2) (\text{now}_i \text{ before}_j - \text{before}_i \text{ now}_j)$ , twice the quantum share's current of Theorem 4, the plain current  $F_{ij}$  per Link's factor being the share's where no stress stands; the form with the plain current holds only where no stress stands, and with a stress it is not conserved (computed, two witnesses; lattice; S.32). (theorem at static paces, weighted; lattice; S.5, S.7, S.32, S.45).*

With the rounding, each Node's remainder adds the term  $(r - r')(\text{next} - \text{before}) / (w p_i^2)$  to the walk of the step's own two-level form  $Q$ , the form of now and before without the wall (theorem, exact in integers, two derivations; lattice; S.5).  $E$  is a diagnostic. Where the paces move it changes exactly in the rationals by  $E(t+1) - E(t) = \mathbf{a}_{\text{next}}^\top \mathbf{D} (w \mathbf{M}(t) - w \mathbf{M}(t+1)) \mathbf{a}_{\text{now}}$ , with the weights  $\mathbf{D}$  of one

interval held and  $w\mathbf{M}$  the numerator's matrix, 0 at paces fixed in time and negative where a well deepens under a standing configuration (a theorem for the identity, by substitution; the sign under a deepening well computed on a standing mode, a lattice reading; lattice; S.45). The momentum density along an axis is the current per axis, and its change is a stress reading on the Links, the momentum flux at uniform coefficients the wave's own stress (theorem for the line, derived for the momentum flux; lattice; S.7, S.45). A plane wave of amplitude  $A$  at the wave number  $k$  has the stress  $T_{aa} = \text{num } A^2 \sin^2 k \geq 0$ , as a wave's pressure is, written into the Link's factor and not the clock (derived; lattice; S.7). That light gravitates by its pressure stands in the list of Section 5.2, derived under the static kernel and untested.

### 3.4 The count is the configuration's quantum share

Three things are called a count. The first is the register's count  $N$ , an integer moved by clicks alone. The second is the quantum share (the field's count in a region, Theorem 3's form over one quantum's norm,  $e_i$  half of Eq. (4)'s Node term,  $E = 2 \sum_i \tilde{e}_i$ , one quantum's norm  $W_c$  in the share's units),  $\sum_i e_i / W_{\text{rec}}$  ( $W_c$  for light), a lattice reading that drifts between clicks by the currents and the remainders within the floors, and where the paces move in time (Section 3.3). The third is the quantum  $W_{\text{rec}}$ , the configuration's energy per quantum read once at its initial data, constant along its passage to the first order in the slow variation of the paces, by the adiabatic invariant (derived, the invariant computed; lattice; S.14). At every Node the configuration's quantum share is

$$e_i = \left[ w(\text{now}_i^2 + \text{before}_i^2) - S_i \text{now}_i \text{before}_i - \text{now}_i \sum_{j \sim i} R_{ij} \text{before}_j \right] \text{div}(2p_i^2), \quad (5)$$

the floor of the rational term  $\tilde{e}_i$  of the conserved form at the Node with its weights; the counting rule reads  $\sum e_i$  in quanta over  $W_{\text{rec}}$  ( $W_c$  for light), rounded half up to the nearest whole. At the vacuum's paces it is exactly  $3 \text{den}(\text{now}_i^2 + \text{before}_i^2) - \text{num now}_i S_6(\text{before}_i)_i$ , with  $S_6$  the six-neighbour sum.

**Theorem 4** (the count is the configuration's quantum share) *Over one step of Rule3 the rational quantum share at a Node changes by exactly the six currents into the Node, each weighted by its Link's factor, plus Rule3's own rounding:  $\tilde{e}_i(\text{next}, \text{now}) - \tilde{e}_i(\text{now}, \text{before}) = \sum_{j \sim i} (q_{ij}^2 / \Gamma^2) F_{ij} - (\text{next}_i - \text{before}_i)(r'_i - r_i) / (2p_i^2)$  (S.5). So at paces fixed in time, over any region the total quantum share changes only by the weighted currents through the region's boundary, read at the levels the step started from, the remainder terms and the floors; where the paces move it drifts with no current crossing any Link (Section 3.3), a diagnostic; the floors each a residual in  $[0, 1)$  per Node and level pair. The total quantum share of any lattice whose Links read the same from both ends and whose every beyond reads 0 (a universe whose massless potential rests at a declared level beyond the open faces lies outside this condition) is non-negative for  $|\text{num}| \leq \text{den}$  at the vacuum's coefficients, with equality for a static uniform level of light and, on an even periodic lattice, for its checkerboard (S.6). One Node's energy can be negative at the initial data's edge, and a region's by at most half a level-square per boundary Port. (theorem under the two conditions; lattice; S.5, S.6).*

Over the whole passage of one forward wave packet the inflow accumulated at a region's boundary equals the region's final energy plus what left through its back, each bounded below by Theorem 4 at the vacuum's paces, where the inflow's coefficients are the identity's and the counting rule's unit is read (in a well the balance carries the paces' weights), so the inflow is non-negative up to the rounding and the counting rule by the quantum shares is well defined (derived from the theorem; clicks; S.5, S.6). Where a counting window is cut inside a passage the NodeDetector credits  $\max(s_R, 0)$ , the floor at 0 the counting rule's declared rule (assumption; clicks). For light, whose rest beyond the faces is 0, the two slits' universe meets the condition; gravity's potential at the declared level 60 is outside it and is not counted.

### 3.5 The Wronskian as the charge

A field of quanta laid as a plane carries its configuration as two level pairs, (re, im), stepped by one call of Rule3 with one set of coefficients, each with its own remainder. The charge density at a Node is the reading

$$W_i = \text{re}_{\text{now}} \text{im}_{\text{before}} - \text{im}_{\text{now}} \text{re}_{\text{before}}, \quad (6)$$

the Wronskian of the two pairs. The line without its division conserves it exactly, with the weights of the conserved form, under the condition of Theorem 3 on the paces (theorem for the conservation; lattice; S.9). The integer step adds to each part the remainder term  $\epsilon = (r - r')/w$ ,  $|\epsilon| < 1$ , so the integer  $W$  changes at a Node, beyond the Wronskian's current through its Ports (S.9), by  $\epsilon_{\text{re}} \text{im}_{\text{now}} - \epsilon_{\text{im}} \text{re}_{\text{now}}$  per interval, not zero in general, one witness computed in S.9 (computed; lattice; S.9). An exactly conserved integer charge is open. The weighted total  $\sum_i W_i/p_i^2$ , the line's without its division, is exact in the rationals at any paces fixed in time. A real configuration has  $W = 0$ , so light, the real configuration of the charge potential, is neutral. A rotating configuration has  $W$  of either sign by its sense, and antimatter, in the model's words, is the configuration of the opposite sense, the same field, gap and inertia with the opposite charge,  $q = \mp Ce$ ,  $C$  the bound state's count in quanta and  $e$  the unit charge, one quantum's Wronskian  $T/2$  (Section 5.1); which sense is nature's matter is the one declared sign (derived for the opposite sense, the sign declared; clicks for antimatter; S.33, S.41, S.52). The charge potential does not read its own level, the declared reads of its row, so light does not read it and a static charge does not bend light (declaration for the reads, an assumption; derived for the bending from it; lattice, no click taken; Table 1's row of the charge potential).

### 3.6 The three exact bands

The rule carries three exact massive bands,  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$ ; this world uses none of them, its matter pair being  $[2, 3]$  ( $[4000, 6000]$  in Table 1). The line of one Node at rest,  $a_{\text{next}} = 2 \cos \omega_0 a_{\text{now}} - a_{\text{before}}$ , is exact in integers only where  $2 \cos \omega_0$  is an integer, so the rule carries exactly three exact massive rotations,  $2 \cos \omega_0 = 1, 0$  and  $-1$ , with the periods 6, 4 and 3 and the pairs  $[1, 2]$ ,  $[0, 1]$  and  $[-1, 2]$  (theorem;

lattice: an integer  $2\cos\omega_0$  strictly inside  $[-2, 2]$  is 1, 0 or  $-1$ ). The rule allows the bands  $[0, 1]$  and  $[-1, 2]$ , and this world does not use them.

## 4 The fields, the bound states and the clusters: what the algebra builds

The algebra builds three things from the one line: fields, each a band of the rule named by its pair; bound states, each a configuration standing in a potential well; and clusters, bound states standing in one another's wells.

### 4.1 Fields, potential fields and the declared universe

A field is a band of the one rule, not a separate law, and its entry in the universe's declaration holds only its pair  $[\text{num}, \text{den}]$  and its shape, the dimension of a field of quanta, 1 for one real line or 2 for a plane, or the sources of a potential field, the form, the form with the three stresses, or the Wronskian, with its divisor  $E_s$  and, for the massless potential, its rest. A field of dimension 3 is declared by no universe's declaration and enters only as a hypothesis under its own name (Online Resource 1, C.5). Under that hypothesis the proton is one quantum of a field of dimension 3, three planes with the rotation among them, one Wronskian per plane into the sign's row, its quarks the three planes; the neutron is three real lines, neutral, no quark field declared, no count per plane and no click of one plane, the conversion is the change between them; the configuration's indivisibility under the click, confinement in the model's words, derived under the hypothesis from the declared row; the baryon number, the flavours, the quark masses and the fractional charges are not derived and not claimed (a hypothesis under its own name; confinement derived under it from the declared row (Online Resource 1, C.5); the neutron's shape this section's declaration under the hypothesis (S.56 (h)); the body's drag of Section 8.1, its three numbers and its  $\alpha_1$ , resting on it and nothing else; clicks).

When a pair is written over the lattice's one denominator  $\Gamma$ , a mass is one integer num, with  $\cos\omega_0 = \text{num}/\Gamma$  and the wall  $6\Gamma^3$ . A particle of nature is read as one quantum of a field, the electron one quantum of a plane matter field, which the rule's own universe does not declare (Table 1), and the proton of its own, and a field's gap, its mass, is declared and not derived (a declaration; Online Resource 1, C.7). What a field carries is set by what Rule3 reads, the clock's pace and the three Link paces, so a potential field carries at most one time part, three axis parts and three odd Link parts,  $1 + 3 + 3$ , the last gravity's potential field's odd lines, sourced by the field's count flux and read by plane configurations alone (derived under the one assumption; lattice; S.56; Section 2.3; the shapes and the reads of this world's fields in Table 1). For a one-part real configuration no  $V$ -proportional read of a current exists that is local, explicit and exact, the staggered step's quantised shift aside, so light and a one-part line are not dragged (theorem for the absent read; derived under the one assumption, untested, for the two predictions; lattice; S.56 (d), (h)).

What general relativity calls curved space stands here as the Link's pace, and what it calls the dragging of frames as the Port's phase; no space is bent in the model

**Table 1** The rule's own universe, its four fields as the universe's declaration holds them (the four fields with their pairs and shapes a declaration; the ranges derived from the pairs, S.11; the odd Link lines derived under the one assumption, S.56; lattice): each field with its pair (in brackets over  $\Gamma = 6,000$ ), what it holds at its divisor, its shape, what it reads, and its free number. A further field, the pair field of light, two real lines never summed, is declared by Bell's declared universe (Section 7)

| Field                                                                                                                | Pair                         | Holds, at the divisor                                                                        | Shape                                                                                                                                | Reads                                                                                          | Its free number                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------|------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| gravity, the time part and the three stresses                                                                        | [1, 1]<br>([6000, 6000])     | the potential level, $E_g$ ( $10^6$ in the two slits' universe, gravity off in all but name) | 1 + 3 + 3 real lines, sourced by the form, the three stresses and the field's count flux                                             | the potential level, its own level among it                                                    | $E_g$ , the hierarchy; its rest $c_{\text{vac}}$ , 60 by declaration                                               |
| the charge potential, whose quanta are light, born by the write                                                      | [1, 1]                       | the sign, 1                                                                                  | 1 real line, sourced by the Wronskian of its readers; under the rotation 1 + 3, three odd axis lines sourced by the sign's current   | the level's potential fields, at the weight 1; never its own level                             | $k$ , the write weight (1 in the rule's own universe; the calibrated value of Section 5.1 in the others), declared |
| the binding potential, not the strong force; its own events have the gap $\cos \omega = \text{num}_b / \text{den}_b$ | [2400, 2401]                 | the potential level, 1 ( $10^6$ in the two slits' universe)                                  | 1 real line, sourced by the form                                                                                                     | the potential level, its own level among it                                                    | the pair, its range $R = 20$ Links following from it (S.11)                                                        |
| matter, the scalar; a quantum that leaves a bound state is a wave packet of matter                                   | [num, den]<br>([4000, 6000]) | nothing                                                                                      | dimension 1, one real line (a charged field of the same pair would be a second entry of dimension 2, a plane; none in this universe) | the level's potential fields, $x = (L_b + L_g) / \Gamma$ ; the plane also the charge potential | $m$ , the mass                                                                                                     |

(derived under the one assumption; lattice; S.2, S.56). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.2.

A potential field's rest falls outside a bound state by  $e^{-\kappa}$  per Link,  $\kappa$  the inverse reach, the field's own band at zero rotation, so the gap is both the mass of the potential's quanta and its inverse reach, Yukawa's form [20], with the range of the list of Section 5.1 (derived; lattice; S.11, S.37). The write of a configuration is laid into two potentials (assumption, fitted, for the pair and the divisor; derived for the range from the pair, S.11; lattice). The first is the binding potential, the gapped pair [2400, 2401] at the divisor 1, its range  $R = 20$  Links following from the pair (S.11); it binds the bound states and is not the strong force. The second is the massless potential [1, 1] at the divisor  $E_g$ , gravity, long-ranged and weak by  $E_g$ , declared large in this universe so that gravity enters no bound state's binding, nature's  $E_g$  read from  $G$  as  $k$  is read from  $\alpha$ .

The massless potential rests at a declared level everywhere, the vacuum level 60 at  $\Gamma = 6,000$  (the vacuum content, the level in the glossary's sense). This world's declaration stands above the floor the law gives, the lattice's own line's floor for a kick, 8 to 13 levels at  $\Gamma = 6,000$  (derived, the kick read on the lattice; lattice; Online Resource 1, B.11). No click bounds it above, since every potential field reads the level and a uniform rest is a change of units (S.12, S.48) (declaration above the floor; derived for the speeds; lattice for the level and its floors, clicks for the speeds; S.12, S.48). Under the law's line the massless potential's own travelling events have light's paces identically in every level, so gravity's events and light through the same wells arrive together,  $\gamma_{\text{GW}} = \gamma_{\text{EM}}$  (the gravitational wave's and the electromagnetic wave's), where GW170817 bounds the difference between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [21] (a comparison). Since the two speeds are one speed, the measurement bounds no rest.

## 4.2 The bound state: the fixed point and its stability

Under the dilation  $\phi_\lambda(x) = \lambda^{-d/2}\phi(x/\lambda)$  the pressure is  $K/\lambda^2$  (a second  $\lambda$ , the dilation's) and each potential's well energy  $V_r/\lambda^{s_r}$ ,  $s_r$  its scaling power at the bound state's size.

**Proposition 1** (the stable bound state, under the dilation and to first order in the well) *For power-law kernels, at a fixed point of the binding potential the virial  $2K = \sum_r \sigma_r s_r V_r$  holds,  $\sigma_r = +1$  for a hollow and  $-1$  for a hill, and the second variation of  $F$  along the dilation is  $F''(1) = \sum_r \sigma_r s_r (2 - s_r) V_r$ . A bound state is stable along the dilation only where this sum is positive; the statement is along the dilation alone, and a full minimum of  $F$  would need the Hessian's mixed block, which no derivation here gives.*

Derived in S.45 by Derrick's scaling [22] and Pohozaev's identity [23] on the lattice, for the continuous functional and to first order in the well, the integer step's version open: in a box the massless potential's well has  $s = 1$ , so  $F = a/R^2 - c/R$  has one critical point,  $R = 2a/c$ , a minimum for every count, Pekar's body [24] (derived; lattice; S.13). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.3.

### 4.3 A bound state's clicks

What a NodeDetector reads of a bound state follows from its configuration and the paces, with no further line. A steady rotation writes no light, and a uniformly moving bound state writes no travelling light in the long-wavelength band (derived; lattice for the Wronskian and the source, clicks for the far NodeDetector's rate; S.40 (b)). A bound state at rest with the plane configuration  $\text{re} = A \cos \omega_b t$ ,  $\text{im} = eA \sin \omega_b t$ ,  $e$  its ellipticity, the ratio of the two parts' amplitudes, has the Wronskian  $W = -eA^2 \sin \omega_b$  at every interval, so the charge potential's source is constant and there is no click. A breathing moves  $W$  between the Nodes, and the light written by the second moment of the charge carries the frequency of the breathing and not the rotation (derived; lattice for the source, clicks for the rate; S.40). The quantum share at a Node is  $(\Gamma^2/p_i^2) A^2 \sin^2 \omega_b / T$ , and under a moving well it drifts with no current through any Link,  $e_i(t_2)/e_i(t_1) = [\sin \omega_b(t_2)/\sin \omega_b(t_1)] [p_i^2(t_1)/p_i^2(t_2)]$  (derived; lattice; S.14). The same identity is Planck's relation on the lattice:

$$\boxed{\text{quantum share at the vacuum's paces} = I \sin \omega, \quad I = A^2 \sin \omega / T}, \quad (7)$$

so  $E = \hbar \omega$  holds, with  $T$  for  $\hbar$  and  $\sin \omega$  in place of  $\omega$ , the invariant standing where nature's number of quanta stands,  $I$  for a one-part real line and  $2I$  for a plane configuration's two parts, and the energy where nature's energy stands (derived; clicks; S.14).

A NodeDetector with Nodes alone credits the energy, a bolometer's reading, Millikan's slope 0. A NodeDetector with a configuration of its own takes one transition per click, the NodeDetector's declaration of Section 2.5 (an assumption), the energy  $T(\sin \omega_j - \sin \omega_i)$ , one photon's  $T \sin \omega_L$  to the order  $\omega_i \omega_j$ , below  $10^{-18}$  at nature's gap, so that a bound state counts photons and reads  $\hbar \omega$  per click, Millikan's slope  $T$  (derived under that declaration; clicks; S.14). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.4.

### 4.4 Clusters: the three kinds of binding, and the cosmology at small redshift

The model writes no line for a cluster: every bound state is bound and configurations may be free, a free configuration gravitating by its form, falling as every field falls and clicking only where it meets a declared NodeDetector's transition, and a cluster is bound states standing in one another's wells (a definition of the law; lattice).

The universe of the model is a finite ball of bound states flying through the static lattice, every one pulled by the inverse square of the others at Kepler's  $G_K$  (Section 5.1). A shell of dust at the radius  $a$  feels the mass inside it, so  $\ddot{a} = -(4\pi G_K/3) \rho a$ ,  $\rho$  the dust's density, and  $\dot{a}^2/a^2 = (8\pi G_K/3) \rho - K/a^2$ , Friedmann's dust equations exactly for the ball's interior [25][26],  $K$  the initial data's declaration; a receding source's redshift is  $1 + z = 1 + \beta$  at first order, Doppler's and Hubble's known forms, the two derivations agreeing exactly (derived, two derivations, under the declared homologous start; clicks; S.29). The paper's cosmology is claimed at  $z \ll 1$  and not beyond: at the second order the ball decelerates, bound or free,  $q_0 = \Omega_m/2 > 0$

( $q_0$  the deceleration parameter,  $\Omega_m$  the matter density parameter; derived under the declared homologous start; clicks; S.29). The supernovae read  $q_0 \approx -0.5$  [27][28][29], a miss named in Section 8. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.5.

## 5 The formulas: the ones shared with nature and the ones the paper adds

Each form below, the vector sector's among them, follows from the rule's coefficients, read by the geometric optics of a packet's band; the form of nature beside it is never an input, except for the three choices of Section 8.2 (the one assumption, line (1) of S.52, the counting rule) and the proposition beside them, the click's twist. The coefficients contain no metric, no Lorentz transformation, no potential and no force, only the paces and the Port's phase: the clock composes by the rule, the Link is the clock twice by the one assumption, and the Port's phase is the potential field's odd line, sourced by the count's flux under the same assumption, with no number put in. For a moving source the potential's flux writes three odd Link levels  $V_{ij}$ , the lattice's  $g_{0i}$ , read as a phase by plane configurations, opposite for the two rotation senses, and by no real one (the one assumption for a moving source; the real line's exclusion a theorem; lattice; Section 2.3; S.56 (d), (h)).  $U = \ell/\Gamma$  denotes the model's potential at the point named, the clock's share per level, which no NodeDetector reads, and  $U_b$  its value at the impact parameter  $b$ .

### 5.1 The formulas shared with nature

A bound state is seen as the model's NodeDetectors see it: by the clicks of the light its parts write and absorb, redshifted by the clock's share and, under the one assumption, bent by  $4U_b$  on the way. The clicks' pattern over time is the fall (Sections 4.2 and 4.3; this section). The forms are stated in two potentials. Every test of gravity takes  $GM$  from Kepler's law of the orbits of matter and  $c$  from light, so it reads  $U_K = G_K M/(c^2 r)$ . In the model a free packet of matter of the pair [num, den] falls in a potential gradient at  $a = c^2 [2 \cos \omega_0 / (1 + \cos \omega_0)] |\nabla U|$ , to first order in  $U$  and at small wave number, the coefficient exact in the gap (derived; clicks; the index lattice; S.18, S.21, S.22, S.28, S.33, S.44). So the orbits of matter measure

$$U_K = \frac{2 \cos \omega_0}{1 + \cos \omega_0} U, \quad 0.800 \text{ at } [2, 3], \quad 1 \text{ as the gap closes} . \quad (8)$$

In  $U$  the composed paces' metric gives, in the post-Newtonian words,  $\gamma = 1$  and  $\beta = 1$  exactly, the metric  $N^2 = e^{-2U}$  and  $h^2 = e^{2U}$ , and the redshift's factor  $f(\omega_0) = 1 + \alpha_U$  with  $\alpha_U = 0.063$  at  $[2, 3]$  (derived under the one assumption; clicks; S.34, S.51). In  $U_K$ , for  $0 < \text{num} \leq \text{den}$ , the same algebra gives  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$  and  $1 + \alpha_K = \tan \omega_0 / \omega_0$ , that is 1.5, 1.25 (the small-gap closed form under the quadratic band; the lattice band's advance is in the list of Section 5.2) and 1.329 at  $[2, 3]$  and 1 as the gap closes (derived under the one assumption; the bending's

measurement its calibration, the perihelion the test of the end's choice, the rest its consequences, the delay and the redshift read against Cassini and Pound and Rebka, the shadow bounded by measurement and not confirmed, conditional on the interior's fate; clicks for  $\gamma$ ,  $\beta$  and the redshift's factor, lattice for the metric's form at the computed pair; S.18, S.19, S.20, S.30, S.31, S.51). The computed pair's values are the lattice's illustration of the band at a coarse gap, and nature's rows stand beside the formulas at nature's pair alone. The bending of light is the case  $\beta = 1$  of the push on a moving configuration, light's own speed;  $\beta$  here is the speed ratio of Eq. (9) and not the post-Newtonian  $\beta$ , and  $\hat{\alpha}$  is the bending angle,

$$\hat{\alpha} = \left| \int \frac{\partial n}{\partial b} dl \right| = 4U_b, \quad \mathbf{F} = \frac{E^2 + P^2}{E} \nabla U = E(1 + \beta^2) \nabla U, \quad \beta = \frac{P}{E} = \frac{v}{c_{\text{local}}} \quad (9)$$

at the small gap, with  $E = \omega$ ,  $P = cN^2 k$  the momentum at the composed paces,  $\beta$  the configuration's speed over the local light speed  $cN^2$  and  $U_b = 3s/(4\pi\Gamma b)$ ,  $s$  the bound state's write per interval over the divisor  $E_g$  (derived under the one assumption; clicks; S.20, S.51). At rest the push is  $E \nabla U$ , for light  $2E \nabla U$ , the factor two being the ruler going with the clock, and at the computed pair the coefficient of  $\beta^2$  read against light's speed is  $\gamma_K - \alpha_K/2 = 1.335$ , against the band's kinetic scale, where the same push has the coefficient 1.0045 (derived under the one assumption; clicks; S.51). For propagation along one lattice axis, a stationary boundary conserves the rotation, so for every field a configuration of wave number  $k$  entering a potential level  $U$  takes the wave number  $k_{\text{in}}$  given by

$$1 - \cos k_{\text{in}} = \frac{1 - \cos k + 3(\text{den} - \text{num})(1 - N^2)/\text{num}}{N^4}, \quad N = e^{-U}, \quad (10)$$

light's index  $k_{\text{in}}/k \rightarrow e^{2U}$  at long wavelength (the optical-mechanical analogy's index), for light exact as  $\sin(k_{\text{in}}/2) = \sin(k/2)/N^2$  with the lattice's chromatic term  $k^2(e^{6U} - e^{2U})/24$  at finite  $k$ , and slow matter's index  $n_m^2 = [1 + 6(\text{den} - \text{num})(1 - N^2)/(\text{num} k^2)]/N^4$ : a slab of potential bends every configuration toward its thick side, the fall of a bound state is the refraction of its own configuration, and a slab thick in the middle is a lens whose image is built click by click (derived, two derivations; lattice for the index, clicks for the focus; S.28). In Universe24 the quantum of action is one click, whose action is the quantum's norm  $T$  and whose energy is  $T \sin \omega$ ,  $T$  at  $\omega = \pi/2$ , the quantum share's ceiling; a bound state whose write is  $s$  quanta per interval over the divisor  $E_g$  makes the rest of the massless potential  $3G_0(r)s$  at  $r$  Links,  $3s/(4\pi r)$  far away, and the level slows the clock by Newton's potential  $\Phi = -c^2 U = -c^2 \ell/\Gamma$ , with Newton's sign, so

$$\frac{G_{\text{clock}}}{c^2} = \frac{3}{4\pi\Gamma E_g}, \text{ that is } G_{\text{clock}} = \frac{1}{4\pi\Gamma E_g}, \quad (11)$$

per quantum of well, in Links and intervals, a reading of  $\Gamma$ , the universe's  $E_g$  and the lattice (derived; lattice for the rest, clicks for the rate; S.25). Kepler's  $G$ , read from the orbits of matter as clicks, is  $[2 \cos \omega_0 / (1 + \cos \omega_0)] G_{\text{clock}}$  by Eq. (8), 0.8 of it at [2, 3] (derived; clicks; S.25). Under the charge potential's rotation the quantum of a field is one unit of the invariant,  $2A^2 \sin \omega_s = T$ , and its Wronskian  $A^2 \sin \omega_s = T/2$  for every field, the unit of charge (definition; lattice; S.26). The law's coupling between two quanta,  $\alpha_{\text{law}}$  (the item of Section 5.2), carries  $k$ , the charge potential's declared write weight (not the wave number). Nature's  $\alpha = 1/137.036$  [30] fixes  $k/(\Gamma E_s)$  and, with the energy line of S.26,  $T/\Gamma^2$ , not  $\Gamma$ , so  $\alpha$  is a coefficient of the declared universe, read from one click as  $G$  is (derived;  $k$  declared; charge universality and the forces' ratio in the small-gap limit, by construction, no test; clicks; S.24, S.25, S.26, S.27, S.42). The force on a bound state of  $C$  quanta is  $\pm C \nabla \theta$ , Coulomb's  $q_1 q_2$  with  $q = \pm C e$  (derived; clicks; S.26 (c)). The two potentials are one kind of field with one kernel, so gravity's coupling between two quanta of the field  $s$  is  $\alpha_g(s) = (3\sqrt{3}/4\pi) \sin^2 \omega_s / (\Gamma E_g)$ . Between two fields the weights cancel,  $(F_e/F_g)_e / (F_e/F_g)_p = (m_p/m_e)^2$  with no number, by construction and no test: nature's  $4.1656 \times 10^{42} / 1.2356 \times 10^{36} = 3.37 \times 10^6$  against  $1836.15^2 = 3.37 \times 10^6$  [30], an identity in the small-gap limit forced by charge universality and  $F_g \propto m^2$  (derived; clicks; S.42).

The units are the electron's: one interval is  $\omega_e \hbar / (m_e c^2) = 1.288 \times 10^{-21} \text{ s} \times \omega_e$  [30] and one Link  $\sqrt{3} \omega_e \bar{\lambda}_e = 6.69 \times 10^{-13} \text{ m} \times \omega_e$  ( $\bar{\lambda}_e$  the electron's reduced Compton wavelength). What a click reads of a quantum is  $T \sin \omega$ , at most  $T$ , against the electron's rest energy  $T \sin \omega_e$ . So LHAASO's 1.4 PeV photon of 2021 [31] bounds  $\gamma_K - 1 = \omega_e^2/2$  below  $6.7 \times 10^{-20}$ , and the Link's bound from the Major Atmospheric Gamma Imaging Cherenkov telescopes' (MAGIC) subluminal dispersion limit [32] gives  $\gamma_K - 1 < 2.0 \times 10^{-28}$  along an axis, under the premise on the mean that S.24 states, the steps through  $\omega_e$  and den in S.24 and S.27 (derived form, bounded by LHAASO and MAGIC and confirmed by neither; clicks; S.24, S.27).  $\Gamma$  reads nothing of  $\omega_e$  or of  $\alpha$ , and no click of the clocks bounds it: the clocks read the mean of the write's dithered sub-unit level, the exact rest, and not  $\Gamma$  (derived, an argument on the mean; the dynamics under a dithered level are open; lattice; S.26, S.27). The gap bound binds the matter field's own denominator.

*The formulas shared with nature, each reached from Rule3 in lattice form, with nature's form beside it, its status and fence (the key of Section 1.3) and its derivation.  $U$  is the clock's share and  $U_K$  the Newtonian potential the orbits measure.*

- The lattice Klein-Gordon equation,  $c_K^2 = \text{num}/(3\text{den})$ ,  $m^2 = 2(1 - \text{num}/\text{den})$ ; Schrödinger's slow limit with  $m^* = 3 \tan \omega_0$ ; nature's form Klein-Gordon; Schrödinger (derived; lattice; S.1, S.34 (c), S.53).
- $E = m^* c_m^2 = \omega_0$ ,  $c_m^2 = \omega_0 / (3 \tan \omega_0)$ ; nature's form  $E = mc^2$  (derived for  $m^*$  from the band,  $c_m$  defined, the identity by construction and no test; clicks; S.18, S.21).
- The conserved form (Theorem 3) and the count as the quantum share (Theorem 4); the Wronskian as the charge; nature's form the conservation of energy and charge (theorem at static paces, for the line without its division; lattice; S.5, S.6, S.9, S.45).
- Poisson's rest  $3s/(4\pi r)$  of the massless potential; the reach  $R = \sqrt{\text{num}/(6(\text{den} - \text{num}))}$  of a gapped one; nature's form Poisson; Yukawa (derived; lattice; S.11, S.37).

- The adiabatic invariant  $A^2 \sin \omega$  and Planck's relation, Eq. (7),  $T$  for  $\hbar$ ,  $\sin \omega$  for  $\omega$ ; nature's form  $E = \hbar \omega$  (derived; clicks; S.14).
- The clock's rate  $1 - f(\omega_0) U$ ,  $f = 2(1 - \cos \omega_0)/(\omega_0 \sin \omega_0)$ , 1.063 at [2, 3]; a cavity of bound states and an atom's beat shift alike by  $N$ ; nature's form the gravitational redshift, Pound and Rebka [33];  $z = (1 + \alpha_K) U_K$ , the same parameter bounded in nature at  $2 \times 10^{-4}$  [12] (derived under the one assumption; clicks; S.15, S.31).
- The bending  $4U_b$ , Eq. (9); against  $U_K$ ,  $(1 + \gamma_K) 2G_K M/(c^2 b)$ ; nature's form Einstein's  $4GM/(c^2 b)$ ; Cassini's  $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$  [34] (derived under the one assumption, which this click fixed (the calibration, no test); clicks; S.20, S.51).
- The delay  $2\sqrt{3} \mu \ln(4x_1 x_2/b^2)$  intervals at  $\gamma = 1$ ,  $\mu = GM/c^2$  in Links,  $x_1$  and  $x_2$  the signal's two ends' distances; nature's form Shapiro's delay [35] (derived under the one assumption; clicks; S.20).
- The perihelion  $6\pi\mu/(a(1 - e^2))$  per orbit,  $a$  the semi-major axis and  $e$  the eccentricity, at the small gap; against  $U_K$ ,  $(2 - \beta_K + 2\gamma_K)/3$  of Einstein's, the lattice band adding its  $k^4$  term at a finite gap (the list of Section 5.2, S.19); nature's form Einstein's advance, the click that fixed the assumption, Mercury's 43 arcseconds per century;  $|\beta - 1| < 10^{-4}$  [12] (derived at the small gap under the one assumption, fixed by the bending's measurement, the perihelion choosing the reading and testing the choice, a test of the choice, one binary choice, and no test of  $\beta$ 's value; the quadratic band an assumption at the computed pair; clicks; S.19, S.34).
- Frame dragging's form for a plane configuration of one rotation sense,  $\delta\omega/\omega_0 = +4 \mathbf{V} \cdot \mathbf{v}/c^2 = 12 \mathbf{V} \cdot \mathbf{v}$  to first order in  $\mathbf{V}$  and for slow matter ( $v_g = \mathbf{k}/m^*$ ), the finite shift  $\omega(k + \theta) - \omega(k)$  exact under the Port's phase and the convention of S.56 (a) (S.56 (e)), with  $\mathbf{V}$  the odd lines' level, the post-Newtonian vector potential of S.56, Einstein's  $+4 \mathbf{V} \cdot \mathbf{p}$  for that sense; the opposite sense  $-4 \mathbf{V} \cdot \mathbf{p}$  and a one-part real line 0, so a body's drag is its plane configurations' signed share of its inertia, which puts the form,  $\alpha_1$ ,  $\alpha_2$  and  $\eta_N$  for bodies among the misses of Section 8.1, where  $\alpha_1 = \alpha_2 = 0$  and  $\eta_N = 0$  hold for a plane configuration of one sense ( $\alpha_2$  and  $\eta_N$  under the relabelling of S.56 (g),  $g_{00}$ 's velocity terms not fixed, S.35) (derived under the one assumption, to first order in  $\mathbf{V}$ , slow matter; clicks; S.56 (e) to (g)).
- The moving clock  $f_v = 1 - v^2/(2c_m^2) - (2\omega_0/\sin 2\omega_0) v^4/(8c_m^4) + O(v^6)$ , Galileo's kinetic form at  $c_m$  to the second order,  $c_m$  a definition, Lorentz's form at the fourth times the band's factor  $2\omega_0/\sin 2\omega_0$  along an axis, direction dependent (S.21), 1 as the gap closes;  $c_m/c = 0.867$  at [2, 3]; nature's form Lorentz's factor; the limiting speeds of matter and light equal: the subluminal case at  $6 \times 10^{-20}$  [36], the superluminal at  $5 \times 10^{-21}$  [37], the earlier bounds at  $10^{-15}$  from photon decay to  $10^{-23}$  for the proton [38] (derived, the second order a definition; clicks, the computed pair's value lattice; S.21).
- De Broglie's  $k = \omega_0 v/c_m^2$  to second order; nature's form  $\lambda = h/p$  (derived; clicks; S.22).
- The fall  $a = \Delta |d\omega_0/d\ell| \nabla \ell$ , universal within a field; between fields  $\eta = 32.3 \omega_e^2$  for beryllium against titanium, below  $1.3 \times 10^{-26}$  at the MAGIC bound and  $4.3 \times 10^{-18}$  at LHAASO's (the torsion balance's  $10^{-13}$  [39]), and  $138 \omega_e^2$  for MICROSCOPE's titanium against platinum, a free-packet estimate; its relation to the bound body's response is undetermined; nature's form Galileo's fall; Eötvös;  $10^{-15}$  for titanium

against platinum [40], the model's  $138\omega_e^2$  for titanium against platinum at natural abundances (135.6 for the single isotopes  $^{48}\text{Ti}$  and  $^{195}\text{Pt}$ ),  $1.8 \times 10^{-17}$  at the bound (derived under the proton's and the neutron's gaps read as two fields, a hypothesis under its own name; a free-packet estimate; its relation to the bound body's response is undetermined; clicks; S.18, S.24, S.44).

- Newton's pull  $1/r^2$  and  $G$  as a reading, Eq. (11); Kepler's  $G$  at 0.8 of the clock's at [2, 3]; nature's form Newton, Kepler (derived; lattice, clicks; S.25).
- The lens, Eq. (10), one formula for light and matter; nature's form  $n = 1 - 2\Phi/c^2$ ;  $n^2 = 1 - 2\Phi/v^2$  (derived, two derivations; lattice, clicks; S.28).
- Coulomb's force over  $\gamma_L$  (the Lorentz factor) between moving charges, 0.995 at  $0.1c$  and 0.866 at  $0.5c$ ; magnetism as the Link's phase; nature's form Coulomb, Lorentz, Peierls' coupling (theorem for the conservation, derived for the force's form under the rotation, not under the plain read; lattice for the identity, clicks for the force; S.33).
- $\alpha = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$ ,  $k$  declared; charge universality  $|q_p| = |q_e|$  by construction; the two forces' ratio  $(m_p/m_e)^2$  between fields; nature's form  $\alpha = 1/137.036$ ; by construction and no test,  $3.37 \times 10^6$  against  $1836.15^2 = 3.37 \times 10^6$  (derived,  $k$  declared; no test; clicks; S.26, S.42). Nature's equality holds to  $10^{-21}$  [41] (a comparison).
- Rabi's rate  $\Omega_R = \epsilon_{ij}/(4\sqrt{\sin\omega_i \sin\omega_j})$ ; the parametric line at  $2\omega_b$  under the pace read alone and none under the rotation (S.41); nature's form Rabi (derived, two derivations (S.16); clicks; S.16, S.41).
- A steady rotation writes no light; a breathing bound state writes at its beat; a bound state counts photons at Millikan's slope  $T$ ; nature's form Larmor's silence of a static charge; Millikan (derived; clicks; S.40, S.14).
- Bohr's levels  $E_n \propto 1/n^2$ ,  $a_0 = \sqrt{3}/(m^*Z\alpha)$ , with the nucleus's angle alone; Landé's  $g_L = 1$ ; nature's form Bohr; the orbital Zeeman effect (derived in the slow limit; line (1) of S.52, an assumption, fitted, made with hydrogen's levels in view, its ground the click's line (a configuration reads every charge potential's row but its own, a meeting being of two quanta and never one; S.52); clicks; S.43).
- Hubble's law  $1+z = 1+\beta$  and Friedmann's dust equations at  $z \ll 1$ ; nature's form Hubble; Friedmann (derived, two derivations; clicks; S.29).
- The two slits' fringe spacing  $\lambda L/d = 16$  Links and the near-field minima at  $y = 15.3$  and  $32.7$ ; nature's form Young's fringes (derived; clicks; S.23); the counts in Section 6.
- Bell's  $E = \cos 2(a-b)$ ,  $S = 2\sqrt{2}$ , Tsirelson's  $2\sqrt{1+\rho^2}$ ,  $\rho$  the pair's ellipticity, GHZ's  $M = -4$ ; every local counting rule gives at most 2; nature's form Bell, Clauser-Horne-Shimony-Holt (CHSH), Tsirelson, Greenberger-Horne-Zeilinger (GHZ) [42] (the counting rule declared and fitted, Bell's, Tsirelson's and GHZ's derived under it, theorem for the local bound; clicks; S.10, S.38, S.47).
- $\gamma_{\text{GW}} = \gamma_{\text{EM}}$  (the gravitational wave's and the electromagnetic wave's), gravity's events at light's speed in every level (derived; clicks; S.12). Nature's form is GW170817's bound between  $-2.6 \times 10^{-7}$  and  $1.2 \times 10^{-6}$  [21] (a comparison).

## 5.2 The formulas the paper adds, and the criterion of refutation

In the model's own potential the bending and the delay are Einstein's to the first post-Newtonian order for a static source and the perihelion is Einstein's at the small gap. The criterion is the one stated in Section 1.1: a derived, click-fenced formula that disagrees with a measurement beyond its stated order refutes the model at that formula, and the refutations found are the misses of Section 8. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.6.

*The formulas the paper adds, each with its number at the computed pair [2, 3] or at nature's, its status and fence, and the click that would read it; "untested" names a prediction no measurement has read.*

- The post-Newtonian parameters with the gap, for  $0 < \text{num} \leq \text{den}$ ,  $\gamma_K = \text{den}/\text{num}$ ,  $\beta_K = (\text{den} + \text{num})/(2\text{num})$ ,  $1 + \alpha_K = \tan \omega_0/\omega_0$ ; at a finite gap the band's  $k^4$  term adds to the advance 13 percent in a coordinate plane, orientation dependent within a percent, and 8 to 10 percent in the (111) plane by the ray integration on the exact band, vanishing as  $\tan^2 \omega_0$  against the Kepler column's advance, and at nature's gap the closed form is exact (S.19); its number 1.5, 1.25, 1.329 at [2, 3]; 1 with the gap (derived under the one assumption; clicks for  $\gamma$ ,  $\beta$  and the redshift's factor, lattice for the metric's form at the computed pair; the click that reads it: Cassini's  $\gamma_K - 1 = (2.1 \pm 2.3) \times 10^{-5}$  bounds  $\text{den}/\text{num} - 1$ ).
- The click's identity under a write (the booking identity),  $\Delta(wQ) = w(x-v)(x-b)$ ,  $Q$  the step's two-level form of now and before (S.46),  $x$  the written level,  $v$  the level Rule3 would write at the Node and  $b$  the level the step leaves, the quantum share's change under any write (S.46); its number exact at the Node at any paces in the rational line, in integers within the remainders' term, with the configuration's own quantum  $W_{\text{rec}}$  (derived; clicks; the click that reads it: every click's write).
- The exponential metric reached from the composed paces,  $N^2 = e^{-2U}$ ,  $h^2 = e^{2U}$ ,  $\gamma = \beta = 1$  in  $U$ ; its spatial second order 2 against the Schwarzschild chart's 1.5 (S.34); its number exact in  $U$ ;  $\alpha_U = 0.063$ , the redshift's factor (derived under the one assumption; lattice, the second order clicks; S.51 for  $\alpha_U$ ; the click that reads it: one click in the deep level).
- The shadow's capture radius  $b = 2em$  against Schwarzschild's  $3\sqrt{3}m$  (S.30); its number  $5.44m$  against  $5.20m$ , 4.6 percent (derived under the one assumption, bounded by measurement and not confirmed; clicks; the click that reads it: the shadow's diameter (Sgr A\* 1.4, M87\* 0.3 standard deviations)).
- The gravity of light, a light packet's integrated write at  $1 \times E/c^2$  under the static kernel, against general relativity's  $2 \times E/c^2$  (S.39); its number the factor 1 (derived under the static kernel (an assumption named), the packet's field open; untested; clicks; the click that reads it: a static bound state's fall toward a light beam (Tolman's pencil), unmeasured).
- Light undragged: a one-part configuration carries no Port phase, so the Lense-Thirring deflection and the Lense-Thirring Sagnac term of light are absent, where every measured drag is of matter; the readings that would read it: a ring laser fixed on the rotating Earth, whose Sagnac frequency carries the Lense-Thirring term in

Einstein's metric and none here, the term GINGER (Gyroscopes in General Relativity) is built to read [43], and a spinning black hole's shadow, asymmetric under Kerr's dragging of light and here symmetric under the spin's reversal whatever the spin, a reading of the Event Horizon Telescope's asymmetry [44] (a Sagnac frequency and a shadow's asymmetry are what a NodeDetector would report, and the paper derives no click formula of them, so the two stand as lattice readings until one is derived; untested) (derived under the one assumption; untested; lattice; S.56 (h)).

- The coupling at speed: the Port's angle is the field's  $\omega_0$  times the delay, so a fast configuration's drag falls as  $1/\gamma_L$ ,  $\gamma_L$  the Lorentz factor, against Einstein's  $+4 \mathbf{V} \cdot \mathbf{p}$  at every speed, where every measured drag is slow; on a ring the two senses of a matter wave split by  $2v_g\theta = 8v_gV\omega_0/c^2$  per Link, and the phase around a closed loop is the circulation of the odd lines' level times  $4\omega_0/c^2$ , independent of  $v_g$ , the matter wave's gravitomagnetic Sagnac phase proper (S.56 (e)), the atom interferometers its readers (derived under the one assumption; untested; lattice; S.56 (h)).
- The lens, Eq. (10), with the chromatic term  $k^2(e^{6U} - e^{2U})/24$  (S.28); its number light's index  $e^{2U}$  and slow matter's  $n_m^2 \propto 1/k^2$  (derived, two derivations; lattice for the index, clicks for the focus; the click that reads it: a chromatic bending of light at finite  $k$ ).
- The fine-structure constant's form,  $\alpha = (3\sqrt{3}/8\pi)k/(\Gamma E_s)$ ,  $k$  declared (S.26); its number  $k/(\Gamma E_s) = 1/28.33$  at nature's  $\alpha$  (derived form, the weight declared; clicks; the click that reads it: the one click that fixes  $k$ ).
- The moving clock's fourth order,  $\omega_0 [(\sum_a n_a^4) \tan \omega_0 + \cot \omega_0]$ , anisotropic by  $\sum_a n_a^4$  (S.21); its number 1.6926 along an axis, 1.2224 along a face diagonal, 1.0657 along the body diagonal at  $[2, 3]$  (derived; clicks; the click that reads it: a clock flown along the lattice's axes against its diagonals).
- Bell's and GHZ's values at rational settings,  $S = 478/169$  and  $M = -4$  (S.10, S.38); its number  $478/169 = 2.82840$  against  $2\sqrt{2} = 2.82843$  (derived under the declared counting rule, which is fitted; an evaluation of the rule, no prediction; clicks; S.10, S.38; the click that reads it: the counting rule's own check).
- The clusters' deceleration  $q_0 = \Omega_m/2 > 0$  at  $z \ll 1$ , bound or free (S.29); its number the sign (derived at  $z \ll 1$  under the declared homologous start; clicks; the click that reads it: the supernovae's Hubble diagram stands against it, a miss of Section 8).
- Compton's factor at a finite gap,  $\lambda_{\text{out}} - \lambda_{\text{in}} = (2\pi/(\sqrt{3}\omega_0))(\omega_0/\tan \omega_0)(1 - \cos \theta)$  in the non-relativistic regime, under the twist proposition of Section 2.5.2; its number the factor  $\omega_0/\tan \omega_0$ , 1 at nature's gap, by construction and no test there (hypothesis, fitted; derived under it, nature's by construction and no test; lattice at the computed pair; S.55; the click that reads it: Compton's shift at a finite gap, nature's at 1).

## 6 The two slits: the fringes from the band and the clicks from the counting rule, the one run their check

The one run this paper reports is the two slits'. The paper contains no experiment against nature: the two slits is one declared universe set against a row written before its run.

### 6.1 The declared universe

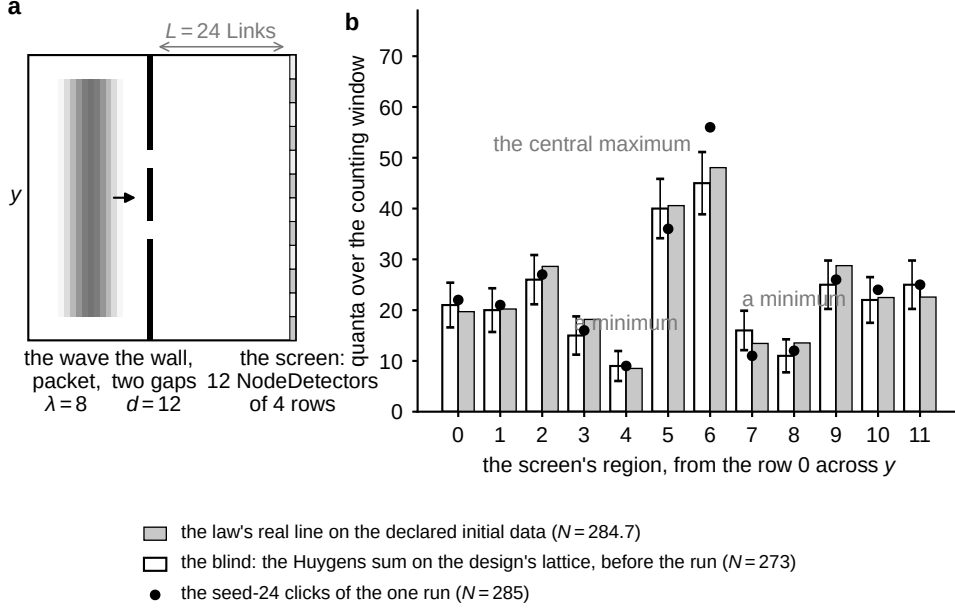
The lattice is flat,  $45 \times 48$  Nodes, its third axis folded, so that the band is exact at  $k_z = 0$  (derived; lattice; S.23). Its faces along  $x$  recede. A packet of light at the wavelength 8 Links,  $k = \pi/4$ , moves along  $+x$ . It is a raised cosine of about ten quanta per Node at its peak over the columns 4 to 16, 1,241 levels at its peak and 1,809 quanta by the integer initial data (1,810.4 by the exact share's sum, S.23); its before level is the band's own exact level, mode by mode, at the vacuum level's paces. Gravity's weight  $E_g$  is  $10^6$  in this universe, chosen so that the packet's own gravity does not focus it, as nature's light never focuses itself (a declaration of the universe; lattice; S.23). The declared universe with its design, its blind row and its reading is deposited with the paper (Statements and Declarations); the design's lattice and both rows stand in the deposit; S.23 holds the geometry and the totals. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.7.

### 6.2 What the model says it must give, and the run

The law's real line is light's line at the vacuum's paces stepped in real arithmetic. On the declared initial data it gives 284.7 units of  $W_c$  through the screen, region by region (computed; clicks; S.23). The line's 284.7 was computed after the run; the one number that stood before the run is the blind row's 273.

The blind row was written before the run on the design's lattice, larger than the declared universe's and with every face open. The blind row's 273 is light's line in real arithmetic propagated on the design's lattice from the two gaps as its sources and summed at the screen over the window, Huygens's construction with the lattice's own line as its propagator, called here the Huygens sum; the line stepped over the whole lattice from the laid packet is a second computation, 284.7 at this universe's exact lay. The difference from 284.7 carries the design's lattice, initial data and window against the declared universe's and Huygens's two sources against the whole lattice's step, undivided here; neither row is a draw, so no scatter enters (computed; clicks; S.23, where both rows' provenance is recorded).

The run gives  $N = 285$  in the region's unit  $W_c$  at both seeds, 24 and 25 (computed, a run; clicks; S.23). That is a bolometer's unit, the energy  $W_c$ , and not a photon count: the screen's 285 units are 674 photons of  $T \sin \omega$  at the run's rotation (S.23), the photon being what a NodeDetector with a configuration of its own would count (Section 4.3). The run's 285 stands within the walk of the line's 284.7, the line at the level 0's paces on the exact initial data and the run at the level 60's; the line at the run's own paces is not computed here (computed; clicks; S.23). The run stands 12



**Fig. 2** The two slits, the one computed universe (computed; the run's  $N$  a run; clicks; S.23). (a) The declared universe. The flat lattice has  $45 \times 48$  Nodes. The wave packet of light, at the wavelength 8 Links, is laid over the columns 4 to 16 (the grey raised cosine) and moves along  $+x$ . The wall at the column 20 has two gaps of three rows, 12 Links apart. The screen at the column 44, 24 Links from the wall, is twelve declared NodeDetectors of four rows each. (b) The screen's row per region over the counting window of the intervals 45 to 130. The design's blind row, the Huygens sum, light's line propagated from the two gaps on the design's lattice, written before the run, is hollow ( $N = 273$ , its bars summing to 275 by the per-Node rounding, with the draw's scatter  $\sqrt{NP(1-P)}$  per region). Beside it, in grey, is the law's real line on the declared initial data ( $N = 284.7$ ). The run gives  $N = 285$  at both seeds. The points are the seed-24 clicks per region of the one run (computed, a run; clicks; S.23)

units over the blind row; it is the check of the line and not the claim. The sum of the absolute differences of the seed-24 clicks from the blind row's counts is 28 of 275 (275 the blind row's sum by its per-Node rounding, 273 its  $N$ ; Fig. 2(b)), a comparison and not a statistical test, since the blind row is the line's and not a draw. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.7.

## 7 Bell and GHZ under the declared counting rule

### 7.1 The declared universe

The counting rule is the one form of the click: the joint energy pairs the parts by their label across the two sides through the source,

$$J(o_A, o_B) = \sum_{\text{window}} \left| \sum_k c_k e_k(o_A) \alpha_k(o_A) e_k(o_B) \beta_k(o_B) \right|^2, \quad c = (1, 1), \quad (12)$$

with  $\alpha_k$  and  $\beta_k$  the amplitude of the part  $k$  at the port on each side,  $e_k(o)$  the port's integer coefficient on the part  $k$ , and  $c_k$  the parts' laid weights, the window's sum standing outside the square. The joint energy is never the square of a local sum, which one NodeDetector could form alone and which gives  $S$  at most 2. One draw is made per pair with the declared seed, the first side from its marginal and the partner from the conditional; each side's write at the Node of its exchange spreads by Rule3 and changes no number of the row, one click per configuration (an assumption, the counting rule's one form; clicks; S.47, S.46). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.8.

## 7.2 What the model says it must give

From the three, with the pair's common phase  $\theta$  uniform over the pairs, A's + port at  $a$  and B's at  $b$ ,  $E$  the correlation of the two sides' signs and  $S$  the CHSH sum,

$$\begin{aligned}
&\text{the meeting:} & P(++) &= \frac{1}{2} \cos^2(a - b), \\
& & E(a, b) &= \cos 2(a - b), \quad S = 2\sqrt{2}; \\
&\text{a local counting rule} \\
&\text{by the energies:} & P(++) &= \frac{1}{4} + \frac{1}{8} \cos 2(a - b), \\
& & E &= \frac{1}{2} \cos 2(a - b), \quad S = \sqrt{2}; \\
&\text{a local counting rule} \\
&\text{by the sign:} & E &= 1 - 4|a - b|/\pi \quad (|a - b| \leq \pi/2), \\
& & S &= 2,
\end{aligned} \tag{13}$$

the second and third lines following by averaging over the pair's phase (derived; clicks; S.10). So  $E(a, b) = \cos 2(a - b)$  under the meeting, exact for the declared pairs by Lagrange's identity of the NodeDetector on equal parts: at the four settings  $E = 119/169, -119/169, 120/169$  and  $120/169$ , and  $S = 478/169 = 2.8284$  exactly, the efficiency 1 by construction and the marginal one half on each side whatever the partner's setting (computed; clicks; S.10).

Without the declaration the lattice gives the second and third lines of Eq. (13), two local counting rules on the same declared universe,  $S = \sqrt{2}$  by the energies of the ports and the sawtooth's 2 by the sign of the larger port, Bell's bound: a pair is one laid configuration, its phase a shared classical variable inside Bell's hidden variable [45], so every local counting rule of it gives  $|S| \leq 2$  [46] (derived under the uniform phase, the numbers computed; theorem for the local bound; clicks; S.10). The same counting rule on four lines gives the three-quantum GHZ correlation [47],  $E_3 = \cos 2(a + b + c)$ , with every pairwise  $E$  equal to 0 and Mermin's  $M = -4$  [48], where every local counting rule gives at most 2 (fitted, the counting rule; derived under it; computed for the numbers; theorem for the local bound; clicks; S.10, S.38, S.47). The detailed paragraphs of this subsection stand in Online Resource 1, Section B.8.

## 8 Limits and what fails

### 8.1 The misses, with their numbers

This section lists the findings against nature, each set beside its measurement. A measurement of nature enters a comparison as what a NodeDetector would report, by the declaration of Section 2.5; whether an instrument of nature is such a NodeDetector is not derived, so a finding against nature is a click-fenced number of the model set beside a measurement read so, and the lattice-fenced comparisons (helium, positronium, the Link's bound) are findings of the lattice and not refutations (assumption, named: nature's instruments read as NodeDetectors; clicks). The derived click-fenced ones refute the model at that formula (derived, the comparisons supporting findings; clicks; lattice for helium, positronium and the Link's bound; S.35, S.50, S.52, S.54). Thirteen stand by name, in the claims list of Online Resource 1; S.50 names its further findings beside them, the gate-joined ions and transmons (S.50 (14)) and the scalar radiation (S.50 (16)) among them, not counted. Ten are set beside a measurement of nature where a line of the law exists: the click-fenced refute the model at their formula, the lattice-fenced (helium's correlation energy and positronium's binding, the Link's bound) are findings of the lattice and not refutations, frame dragging for bodies and the  $\times$  polarisation among the click-fenced, derived under the static read, which the odd write keeps (S.50 (1); S.56 (d)). Three are places with no line, the neutron's lifetime, Pauli's exclusion and the loops, where nothing is derived and nothing contradicted and the row names what a line would have to give (the key of Section 1.3; the claims list of Online Resource 1).

Frame dragging's form,  $+4 \mathbf{V} \cdot \mathbf{p}$ , is derived for a plane configuration of one rotation sense; the opposite sense is  $-4 \mathbf{V} \cdot \mathbf{p}$  and a one-part real line 0, so a body is dragged by its plane configurations' signed share of its inertia,  $Z/A$  for nature's matter under the proton of Section 4.1: Gravity Probe B's rotors (quartz, 0.50)  $19.6$  against  $37.2 \pm 7.2$  milliarcseconds per year [49], the LAGEOS and LARES satellites (0.46 and 0.40) against the Lense-Thirring tests that read general relativity's value, LAGEOS first [50] and the three satellites together within a few percent [51], a finding against nature by name; with it the frame-dragging parameter a body of share  $f$  reads is  $\alpha_1 = 8f - 8$ ,  $-4$  at quartz's  $Z/A = 0.50$  and  $-4.3$  to  $-4.8$  at the satellites' 0.46 and 0.40, against the binary pulsar's bound  $|\alpha_1| < 4 \times 10^{-5}$  on the one post-Newtonian  $\alpha_1$  [52], a miss at every share below 1; antimatter anti-dragged and positronium undragged, predictions no measurement has read (derived under the one assumption for a plane configuration of one sense; the body's share  $Z/A$  and its numbers under the proton hypothesis of Section 4.1 as well; clicks, the two predictions lattice; S.56 (e) to (h)). The miss is a bound body's: a plane configuration of one sense is dragged at Einstein's first-order form at nature's gap and a one-part real line not at all, so what the readings test is the law's write and the proton hypothesis of Section 4.1 together and not either alone; the neutron, three real lines under the hypothesis, is undragged, and a reading of its drag beside a rotating mass would tell whether the model computes a bound body's drag (derived for the two configurations; the neutron's drag a prediction no measurement has read; clicks; S.56 (d), (e), (h)). Under the relabelling that gives a plane configuration of one sense its  $\alpha_2 = 0$  (S.56 (g)), a body of share  $f$  reads

$\alpha_2 = f - 1$  and  $\eta_N = 4\beta - \gamma - 3 - \alpha_1 + 2\alpha_2/3 = (22/3)(1 - f)$  at  $\beta = \gamma = 1$ ,  $-0.5$  and  $3.7$  at one half; with the odd lines read as  $V_j$  alone and no relabelling,  $\alpha_2 = -1$  at every share and  $\eta_N = 3.3$  at one half; the split between the two vector potentials is a gauge the law does not fix, since it needs  $g_{00}$ 's velocity-dependent terms (S.35), so the two are computed under their readings and not derived, where  $\alpha_1 = 8f - 8$  is; against the Sun's  $|\alpha_2| < 4 \times 10^{-7}$  [53] and lunar laser ranging's  $|\eta_N| < 4.4 \times 10^{-4}$  [12] the miss stands under either reading and at every share below 1 (computed under the gauge named, not derived; clicks; S.56 (g), S.35).

The detailed paragraphs of this subsection stand in Online Resource 1, Section B.9.

## 8.2 What is put in

The model declares and does not derive the following items, each a declaration of the universe (assumption for every declaration; lattice, clicks; the claims list of the supplement). Three choices were made with a known result in view. They are inputs and not predictions, beside the universe's fitted numbers named below. The first is the one assumption  $\hbar N = 1$ , for a source at rest and for one that moves, the odd write of Section 2.3, fixed by the bending's measurement and tested by the perihelion's, which chose its end (Section 2.3). The second is the atom's levels, line (1) of S.52, an assumption, fitted, made with hydrogen's levels in view, its ground the click's line (a configuration reads every charge potential's row but its own, a meeting being of two quanta and never one; S.52). The third is the counting rule, a click's energy as the square of the meeting, chosen to reach nature's correlations (Section 7). One line beyond the law stands beside them, the click's twist of Section 2.5.2, chosen with the per-click conservation of momentum in view, an input to no claim of the paper, the conditional results of Section 2.5.2 its own, Compton's factor its one lattice term under it, 1 at nature's gap, by construction and no test (assumption, fitted, for  $\hbar N = 1$ ; lattice; the counting rule's one form an assumption of the law, fitted, adopted with nature's correlations in view, the square Born's declaration; clicks; the click's twist a hypothesis, fitted; the claims list of Online Resource 1). The free rest's computability is a hypothesis under its own name and no claim of this paper. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.10.

## 8.3 What a world closer to nature needs

This world differs from nature's along two axes. The first is the box's size, which behaves as open beyond its light-crossing (Online Resource 1, Section B.11). The second is  $\Gamma$ 's: the discreteness' marks shrink as  $\Gamma^{-2/3}$ , so a world closer to nature is a larger  $\Gamma$  with the one-way face, not a larger count of Nodes; the calculations such a world needs are named and not done here. The detailed paragraphs of this subsection stand in Online Resource 1, Section B.11.

## 9 Conclusion

One line of integer arithmetic, Rule3, gives in lattice form the forms of the list of Section 5.1, each under the condition its item states, and the two slits' row computed

at one declared universe. Beside the line stand the declarations named with it and no number of nature but the two measurements of the one assumption, the bending's, which fixed it, and the perihelion's, which chose the reading and tests the choice, the atom's line (1) and the correlations the counting rule was chosen for, with the universe's calibrations (Section 8.2). Born's rule is not recovered but put in, with Bell's  $2\sqrt{2}$  its evaluation. Among those forms the vector sector of gravity is the paper's own result, and read on bodies it is a finding against nature. The one assumption, which clocks a Link twice, applied to a moving source gives the Port's phase with no number put in, and with it frame dragging's form,  $g_{0j} = -4V_j$ , for a plane configuration of one rotation sense; the drag is odd in the sense, so a body is dragged by its plane configurations' signed share of its inertia, about one half for nature's matter under Section 4.1's proton hypothesis, 19.6 against Gravity Probe B's  $37.2 \pm 7.2$  milliarc-seconds per year [49] and 0.46 and 0.40 of the Lense-Thirring satellite tests' value [50, 51], with  $\alpha_1 = 8f - 8$  for a body of share  $f$ ,  $-4$  at one half, against the binary pulsar's bound on the one post-Newtonian  $\alpha_1$  [52];  $\alpha_1 = \alpha_2 = 0$  and  $\eta_N = 0$  hold for a plane configuration of one sense ( $\alpha_2$  and  $\eta_N$  under the relabelling of S.56 (g),  $g_{00}$ 's velocity terms not fixed, S.35) and not for the bodies the bounds are read on (the Sun's and the solitary pulsars' on  $\alpha_2$  [53][54], lunar laser ranging's on  $\eta_N$  [12]), where the write without the odd Link part gave  $\alpha_1 = -8$  and no drag (derived under the one assumption and the band's own Lorentz form at long wavelength and small gap, S.21, to first order in  $\mathbf{V}$ , slow matter; the body's share  $Z/A$  under the proton hypothesis of Section 4.1; clicks at nature's gap; Section 8.1; S.56 (e) to (g)). Its two lattice terms stand as predictions, light undragged by a rotating mass, which a ring laser's Lense-Thirring term [43] and a spinning black hole's shadow [44] can read, and a fast configuration's coupling falling as  $1/\gamma_L$  (derived under the one assumption; lattice; the two predictions untested; S.56 (h)).

The contribution is the explicit chain from the rule through the declarations and the derivations to the clicks, every input and every miss named, so that each step can be checked and each reading repeated. For each field the lists touch, the weak field, the atom's levels, the two slits and Bell's pairs, the cosmology at small redshift, the paper gives one handle for that field's own research: the formula in lattice form with its status and its fence, to set beside the field's measurements or to refute at it by the criterion of Section 1.1. The paper has one claim, that Universe24 helps compute things in nature, and none of it is a proof about nature. The model is a wave mechanics of configurations on the Nodes and their meetings. What is put in is named (Section 8.2). The form of a click is claimed and its cause is not. Why nature is so, we do not claim to know.

The conjectures stand in Online Resource 1, Section B.12, one line each (inspiration for the directions, hypotheses under their own names; the click's twist a hypothesis, fitted, of Section 2.5.2, its Compton factor derived under it; no number of the paper's claims rests on the twist or on the click hypothesis; the nuclear misses and the Fermi constant  $G_F$ 's of Section 8.1 stand under the nuclear and the conversion hypotheses by name; the conditional results of Section 2.5.2 the twist's own; clicks for the hypotheses, lattice for the twist's Compton factor at the computed pair, nature's by construction

and no test; Online Resource 1, C.5, C.6 and C.7). The detailed paragraphs of this section stand in Online Resource 1, Section B.12.

The paper gives the direction and not the whole, so that others can take the method, with this set of rules or another, and continue.

## Statements and Declarations

**Funding.** None. **Competing interests.** None. **Ethics approval and consent.** Not applicable. **Data availability.** The law’s document and the two slits’ universe with its design, its blind row and its reading are in the Git repository <https://github.com/Closer24/Universe24> (private until the article’s arXiv posting, then public), archived at Zenodo under the concept DOI 10.5281/zenodo.22738745 [55], this version’s deposit with this submission, MIT licence. **Code availability.** The repository also holds an implementation for building worlds, frozen before the odd write of Section 2.3; the two slits’ run of Section 6 is its one run the paper reports, at the state the run’s record names, on a universe with gravity off in all but name, so the run is the law’s line as Section 2.2 states it, nothing of Section 5.1’s vector sector rests on it, and no other claim of the paper does. **Author contributions.** The author conceived the model, carried out the derivations and the computations with the tool stated below, and wrote the paper. **The use of a large language model.** The author used a large language model (Claude, Anthropic, through Claude Code, 2026) as a tool under the author’s direction for the computations, the derivations and their checks and the drafting of the text; the model is not an author, and the author reviewed every statement and takes full responsibility for the content.

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